



## Research Paper

# Economic inequality, social capital and the role of merit and luck: An experimental study

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## ABSTRACT

Research consistently links inequality to lower social capital, the networks, norms, and trust that facilitate cooperation. Yet not all inequality is perceived the same: people are generally more accepting of inequality arising from merit than from luck. This study investigates whether the negative effect of unequal environments on social capital depends on the source of inequality: merit or luck. Using trust, prisoner's dilemma, and stag-hunt games, we find that inequality, regardless of its source, reduces trustworthiness and undermines coordination to the payoff-dominant outcome in the stag-hunt game. Notably, luck-based inequality results in higher trust compared to merit-based inequality, even though participants perceive meritocratic allocations as fairer.

## 1. Introduction

Economic inequality is a key concern for academics, policymakers, and the general public alike (Champernowne, 1973; Pickett and Wilkinson, 2010, 2019; Piketty, 2014, 2020; Stiglitz, 2012). The World Inequality Report (Chancel et al., 2022) reveals that the combined income of the bottom half of the global population falls short of those of the top 0.1 per cent. The wealth gap is even more striking, with the richest 0.1 per cent owning nearly ten times as much as the poorest half. Empirical research consistently links high inequality to adverse outcomes like poverty (Marrero and Servén, 2022), crime (e.g., Nivette, 2011; Stucky et al., 2016), mental health issues (Ribeiro et al., 2017), worse heart health (Dewan et al., 2019), and lower life expectancy (Babones, 2008; Pickett and Wilkinson, 2015) among others.

Beyond its direct connection with material well-being, economic inequality has the potential to undermine broader societal constructs like social capital: the networks, norms, and trust that enable individuals to act together more effectively to pursue shared objectives (Putnam, 1995, p. 67). Studies consistently find an inverse correlation between inequality and social capital as measured by trust (e.g., Barone and Mocetti, 2016; Kawachi et al., 1997; Knack and Keefer, 1997; Ram, 2013) or voluntary contributions to public

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goods (e.g., Cherry et al., 2002; Hargreaves Heap et al., 2015). Strong social capital offers significant economic benefits, such as fostering economic growth, improving investment outcomes, and enhancing governmental efficiency (Knack and Keefer, 1997; Whiteley, 2000). Moreover, social capital plays a crucial role in mediating the relationship between inequality and various adverse social and health outcomes (Delhey and Dragolov, 2014; Elgar, 2010; He et al., 2021; Kragten and Rözer, 2017; Layte, 2012; Layte et al., 2019; Mackenbach et al., 2016; Rözer et al., 2016).

A separate strand of research explores the role of fairness perceptions in shaping people's acceptance of inequality. Inequalities arising from merit owing to differences in effort or performance, are perceived as fairer and, thus, more acceptable than those stemming from chance (Almás et al., 2020; Cappelen et al., 2013, 2020; Fong, 2001; Konow, 2000; Starmans et al., 2017). Yet it remains unclear whether these differences in attitudes translate into different behaviours that shape social capital. If individuals knowingly operate in a meritocratic environment rather than one where outcomes are driven by luck, does it affect their willingness to engage in activities that build social capital? Answering this question is crucial for understanding how perceptions of fairness moderate the effects of inequality on social capital and whether differences in perceptions necessarily translate into differences in behaviour.

This study aims to tackle two key questions. First, how does inequality, regardless of its source, shape behaviour in ways that affect social capital? Second, does the source of inequality, whether based on merit or luck, alter these effects? Establishing causal relationships in this context requires identifying exogenous variation in both inequality levels and its underlying sources, which is not always feasible with observational data. To address this limitation, we conduct an online experiment that allows us to isolate the causal effects of inequality and its source on trust, cooperation, and coordination, key elements of social capital.

We structure our experimental design around treatment groups with either an equal or unequal distribution of initial endowments. In the unequal treatments, participants are split in two categories: those with lower endowments (50 tokens) and those with higher endowments (100 tokens). For clarity, we refer to these groups as "poor" and "rich," respectively, though these labels are used solely for expositional ease. The experimental instructions refer only to numerical endowment levels, not to these emotive terms. Participants therefore know only whether they are allocated 50 or 100 tokens, without any reference to being "rich" or "poor."

The key variation lies in how these inequalities are determined. In the merit-based unequal condition, participants earn their endowments through performance on a real-effort task and know this to be the case. In the luck-based unequal condition, participants complete the same real-effort task, but are informed that endowments are randomly assigned and unrelated to their performance. To establish benchmarks, we include two equal-endowment control treatments. In one, all participants receive the same high endowment (100 tokens) as the *rich* in the unequal treatments; in the other, all receive the same low endowment (50 tokens) as the *poor*. Control group participants complete the same real-effort task but are informed that everyone in their treatment receive the same endowment independent of task performance. All participants know how endowments are allocated in their treatment group but are blind to other treatments.

After participants are allocated to the four treatment groups (i.e., high-endowment equal, low-endowment equal, merit-based unequal or luck-based unequal), they participate in three paired economic games. These include the trust game (Berg et al., 1995), which is essentially a sequential move variant of the prisoner's dilemma game; and two simultaneous move games, the stag-hunt game (Cooper et al., 1992) and the prisoner's dilemma. These games serve to measure core dimensions of social capital: trust, trustworthiness, coordination, and cooperation (Putnam et al., 1994). While in the equal treatments, pairings occur exclusively between participants with symmetric endowments (i.e., *rich* paired with *rich* and *poor* with *poor*), the unequal treatments allow for both symmetric and asymmetric pairings (i.e., *rich* paired with *poor* and vice versa).

Our analytical strategy proceeds in two stages. First, we examine how inequality, per se, affects behaviour by comparing outcomes between the equal and unequal treatments. Second, we explore differences in behaviour between the two unequal treatments depending on whether the inequality arises from merit or luck. Crucially, in all cross-treatment comparisons, we hold pairings (and therefore endowments) constant to isolate the treatment effect.

Consider the first comparison between equal and unequal treatments. In an equal society, a poor individual interacts solely with other poor individuals in a uniformly poor environment with no inequality. In an unequal society, the same interaction occurs, but within a society that also includes contact with rich individuals, making inequality salient. Would behaviour change despite the identical poor-poor pairing across these two societies? We pose the same question for the rich: does a rich-rich interaction lead to different outcomes when situated in an unequal versus an equal society? By holding pairings fixed and varying only whether the relevant society is equal or unequal, we avoid any potential confounds and isolate the effect of inequality alone. We restrict our analysis to symmetric "*rich-rich*" and "*poor-poor*" pairings only because equal societies, by design, are either uniformly rich or uniformly poor and thus contain no asymmetric pairs (*rich-poor* or *poor-rich*). We pool results from both the merit and luck-based unequal treatments, and compare these with the equal treatments, holding pairings (either *rich-rich* or *poor-poor*) constant. We find that in the unequal treatments, even when participants with identical endowments interact, they display lower trustworthiness in the trust game and reduced coordination to the payoff-dominant equilibrium in the stag-hunt game. This pattern holds for both the *rich* and the *poor* pairs in the stag-hunt game but in the trust game, the decline in trustworthiness is evident predominantly among the *poor* pairs.

In our second comparison, we examine whether the source of inequality, merit or luck, shape behaviour in meaningful ways. Here, we analyse all four pairing types (i.e., *rich-rich*, *rich-poor*, *poor-rich* and *poor-poor*) and ask whether behaviour varies between merit-based and luck-based unequal societies, holding the pairing type constant. While prior experimental studies document how unfair inequality affects trust (e.g., D. Fehr, Rau, et al., 2020; Rodrigo-González et al., 2021), disentangling the effects of inequality from perceptions of unfairness remains challenging, as these factors are often conflated. By isolating the source of inequality while keeping the degree of inequality unchanged, our study provides a novel perspective. We find that, overall, the source of inequality has limited influence on behaviour except that trust levels are higher under luck than merit, especially in *rich-poor* pairings. This finding is striking as participants consistently rate merit and egalitarian systems as fairer than luck; yet perceived fairness does not translate into greater

trust, revealing a disconnect between fairness perceptions and trusting behaviour. One possible explanation is that merit-based inequality fosters a stronger sense of entitlement. We elaborate on this later after presenting our results.

Our findings contribute to the broader literature on inequality and social capital by demonstrating that while merit-based inequality is often perceived as fairer than luck-based inequality, it does not necessarily foster stronger trust. Our results further reveal that both merit-based and luck-based inequalities, when taken together, weaken trustworthiness and coordination, reinforcing the need for policies that mitigate inequality to support long-term economic stability and social cohesion. The paper proceeds as follows. Section 2 presents a selective literature review and the hypotheses motivated by it; Section 3 details the experimental design and procedures; Section 4 presents the results of our experiment; and Section 5 offers concluding remarks.

## 2. Literature review and research hypotheses

We begin by exploring current research on the relationship between inequality and social capital. Next, we look at the literature on the impact of the source of inequality on people's fairness preferences and social capital. On the basis of this review, we generate our testable hypotheses.

### 2.1. Relationship between inequality and social capital

Putnam (1995, p. 67) defines social capital as “features of social organisation, such as networks, norms and social trust that facilitate coordination and cooperation for mutual benefit.” Trust and cooperation are essential pillars of social capital, with a reciprocal relationship: trust fosters cooperation, and cooperation reinforces trust (Putnam, Leonardi, and Nanetti 1994, p. 171). Evidence indicates that social capital acts as a key channel through which inequality affects individual health and well-being: higher inequality tends to erode trust, which in turn diminish health and life satisfaction (Delhey and Dragolov, 2014; Elgar, 2010; He et al., 2021; Kragten and Rözer, 2017; Layte, 2012; Layte et al., 2019; Mackenbach et al., 2016; Rözer et al., 2016). Additionally, robust social capital contributes to economic benefits by driving growth, enhancing investment returns, and improving government efficiency (Knack and Keefer, 1997; Whiteley, 2000). Understanding how inequality affects social capital is, therefore, crucial.

Empirical studies consistently show a negative correlation between income inequality and trust, a key component of social capital (e.g., Barone and Mocetti, 2016; Kawachi et al., 1997; Knack and Keefer, 1997; Ram, 2013). Yang and Konrath's (2023) meta-analysis, synthesising 100 studies with over 2.5 million observations, confirms a weak but significant negative correlation between inequality and prosociality, often used as a proxy for social capital. However, using observational data to establish causality and detect its direction is challenging as the variables cannot be randomly assigned, making it difficult to rule out biases like confounding, reverse causality, omitted variables or simultaneity (Angrist and Pischke, 2010). Moreover, most empirical studies that rely on survey questions to measure trust are often criticised for their ambiguity and failure to capture actual behaviour (Glaeser et al., 2000).<sup>1</sup> Economic experiments address these issues by randomising key variables and using salient incentives to measure trust as a behaviour, allowing for more accurate causal inferences (Camerer, 2015; Camerer and Hogarth, 1999; Chaudhuri, 2009; Falk and Heckman, 2009).

Studies using the trust game (Berg et al., 1995) have been central to experimentally examining how inequality impacts trust and trustworthiness, key components of social capital. Anderson et al. (2006) found that trust declined when participants were privately aware of income inequality in the form of asymmetric show-up payments, though trustworthiness remained unaffected. In a repeated trust game with cumulative wealth, Greiner et al. (2012) showed that inequality led to lower initial trust compared to equal societies, where trust started higher but gradually declined over time. Hargreaves Heap et al. (2013) explored the effects of endowment inequality on trust in both the trust game and the gift-exchange game (E. Fehr et al., 1993), finding that inequality, especially when income differences were visible, undermined trustworthiness in both games, and reduced trust more significantly in the labour market setting. D. Fehr et al. (2020) similarly found that unfair inequality, created through performance-based payments that rewarded winners and made it harder for others to catch up, diminished both trust and trustworthiness. Xiao and Bicchieri (2010) examined how inequality aversion affects reciprocity in the trust game. Their results showed that second movers display lower trustworthiness when initial endowments were asymmetric, suggesting that the desire for equality can outweigh tendencies toward reciprocity.

While most prior research has focused on trust and trustworthiness, cooperation in social dilemma games, where individuals must choose between the common good and their self-interest, also serves as a useful proxy. In this study, we look at two measures of cooperation: the prisoner's dilemma game and a stag-hunt game. The games differ in their equilibrium predictions: while the prisoner's dilemma has a unique Nash equilibrium where both players defect, the stag-hunt gives rise to two possible equilibria: one payoff dominant and the other secure. However, the games are similar in respect to what is important for our purposes. Mutual cooperation is socially optimal in both games, but it requires that a player must trust that the other will also cooperate (Skyrms, 2003, p. 3). Further, Rabin (1993) suggests that with reciprocal preferences, a prisoner's dilemma resembles a stag-hunt game. Skyrms (2001, 2003) argues that many social processes have the features of a stag-hunt game where the key issue is coordinating to the payoff dominant outcome, which requires mutual trust among players.

Studies have shown that cooperation is more challenging in asymmetric prisoner's dilemma games, where payoffs are unequal,

<sup>1</sup> For example, the General Social Survey uses the question, “Generally speaking, would you say that most people can be trusted or that you can't be too careful in dealing with people?” to measure trust. Researchers have argued that this question can be vague and difficult to interpret (Glaeser et al., 2000).

compared to symmetric scenarios with equal payoffs (Ahn et al., 2007; Sheposh and Gallo, 1973).<sup>2</sup> However, these studies compare scenarios with different payoff matrices, making it difficult to disentangle whether observed behavioural changes result from inequality per se or from altered incentives. To our knowledge, there has been limited exploration of how behaviour varies when the payoff structure itself remains constant, but the context generating inequality changes, for instance, whether inequalities stem from meritocratic or luck-based factors. This omission leaves a gap in our understanding of how contextual framing of economic inequality shapes cooperative behaviour. Moreover, this gap is even more pronounced in the context of stag-hunt games, where inequality remains an underexplored dimension. The above discussion brings us to our first hypothesis:

**Hypothesis 1.** Holding pairings (and therefore endowments) constant, inequality erodes social capital relative to equality. Specifically, compared to equal societies in which all participants receive identical endowments (either 50 or 100 tokens), in unequal societies (regardless of whether the inequality arises from luck or merit), we expect:

- (a) lower levels of (i) trust and (ii) trustworthiness in the trust game.
- (b) lower rates of coordination to the payoff dominant equilibrium in the stag-hunt game, and
- (c) lower rates of cooperation (implying higher rates of defection) in the prisoner's dilemma game.

## 2.2. Source of inequality, fairness and inequality tolerance

Starmans et al. (2017) highlight a discrepancy between laboratory data showing evidence of inequity aversion (Bolton and Ockenfels, 2000; E. Fehr and Schmidt, 1999) and field data that suggests people prefer inequality if it is perceived as fair. In typical lab designs, participants begin with equal endowments, which makes unequal outcomes appear unfair. Yet, in real-world settings, fairness judgments hinge more on merit, leading individuals to accept, and even defend, inequality that stems from effort rather than luck. This 'unfairness aversion' explains why individuals accept inequality and resist redistribution when wealth differences are attributed to merit rather than luck (Alesina and Angeletos, 2005; Alesina and Giuliano, 2011; Alesina and La Ferrara, 2005; Fong, 2001; Konow, 2000). Numerous experiments reinforce this merit-based view, showing a pronounced preference for inequality arising from effort rather than chance (Almås et al., 2020; Cappelen et al., 2007, 2013, 2022, 2023; Rustichini and Vostroknutov, 2014).

Most of these studies measure inequality tolerance by examining how willing individuals are to redistribute earnings based on whether observed income gaps emerge from merit or luck. Their findings show that people are relatively unbothered by inequalities that reflect genuine effort, yet they become more inclined to redistribute when luck drives those disparities (Almås et al., 2020; Cappelen et al., 2013, 2022, 2023). This preference for effort-based inequality may have evolutionary origins: research on children and non-human primates demonstrates that they, too, favour rewarding effort rather than simply dividing resources equally (Baumard et al., 2012; Brosnan and de Waal, 2003, 2014; Dindo and De Waal, 2007; Hamann et al., 2014; Kanngiesser and Warneken, 2012).

We argue that it is crucial to distinguish between social capital (encompassing aspects like trust, cooperation, and prosociality) and the willingness to accept inequality, as captured by redistribution choices. Trust and cooperative behaviour manifest in contexts far beyond redistributing income, such as when individuals share information, lend resources, or collaborate on group tasks. Although it is intuitive to expect that people would be more trusting under systems they perceive as fairer, this is not necessarily the case: one may accept merit-based pay disparities as fair, while still harbouring distrust or reluctance to help. An aversion to unfair inequality helps us understand when people object to certain income gaps, yet it does not fully capture how trusting they are toward one another. Thus, even though the broader literature on inequality acceptance shows how the source of inequality (luck versus effort) shapes fairness judgments and influences the acceptance of disparities, they leave an important question unanswered: how do these perceptions affect broader societal outcomes such as trust and cooperation?

D. Fehr et al. (2020) examined the impact of unfair inequality on trust and trustworthiness using a two-stage experiment. In the first stage, participants took part in a real-effort task under two different payment schemes: piece-rate and relative-payment, where the latter introduces elements of both inequality and unfairness through competitive pay and time bonuses that advantage early winners. In the second stage, participants played the Berg et al. (1995) two-player trust game. They found that the relative-payment scheme, which gave rise to both higher inequality and perceived unfairness, significantly reduced trust and trustworthiness compared to the piece-rate scheme, suggesting that unfair inequality erodes social capital. Similarly, Rodrigo-González et al. (2021) found that trust, though not trustworthiness, was lower when initial endowments were based on effort (a setup that created inequality), than in a baseline treatment where everyone received the same endowment. Further, in the effort-based treatment, participants were more trusting of those with higher endowments, suggesting that they regarded past effort as a signal of reliability.

While both studies suggest that unfair inequality undermines key aspects of social capital, they face challenges in fully isolating the effects of unfairness from inequality, as their treatments produced both unequal and unfair outcomes. Our study addresses this gap by comparing a fair, merit-based system with an unfair, luck-based system, while keeping inequality levels constant. Moreover, rather than examining how individuals behave when interacting with others of similar or different wealth, we focus on the broader effects of being embedded in unequal environments. Everyone interacts under the same pairing conditions, but the origin of inequality (merit

<sup>2</sup> The asymmetric prisoner's dilemma modifies the standard prisoner's dilemma payoff matrix so that one player can earn more than the other, even when both cooperate. This asymmetry enables researchers to identify cooperation and defection patterns and predict how players behave under unequal conditions.

versus luck) varies across treatments. Finally, we extend our analysis beyond the trust game to the prisoner's dilemma and the stag-hunt game, with each participant engaging in all three games, to measure the impact of the source of inequality on both social trust and cooperation. This line of inquiry informs our second hypothesis:

**Hypothesis 2.** Holding pairings (and therefore endowments) constant, social capital is stronger under merit-based inequality than under luck-based inequality. Specifically, compared to the luck-based unequal condition, in the merit-based unequal condition, all relevant pairs of participants, will exhibit:

- (a) higher levels of (i) trust and (ii) trustworthiness in the trust game,
- (b) higher rates of coordination to the payoff dominant equilibrium in the stag-hunt game, and
- (c) higher rates of cooperation (implying lower rates of defection in the prisoner's dilemma game).

### 3. Experimental design and procedures

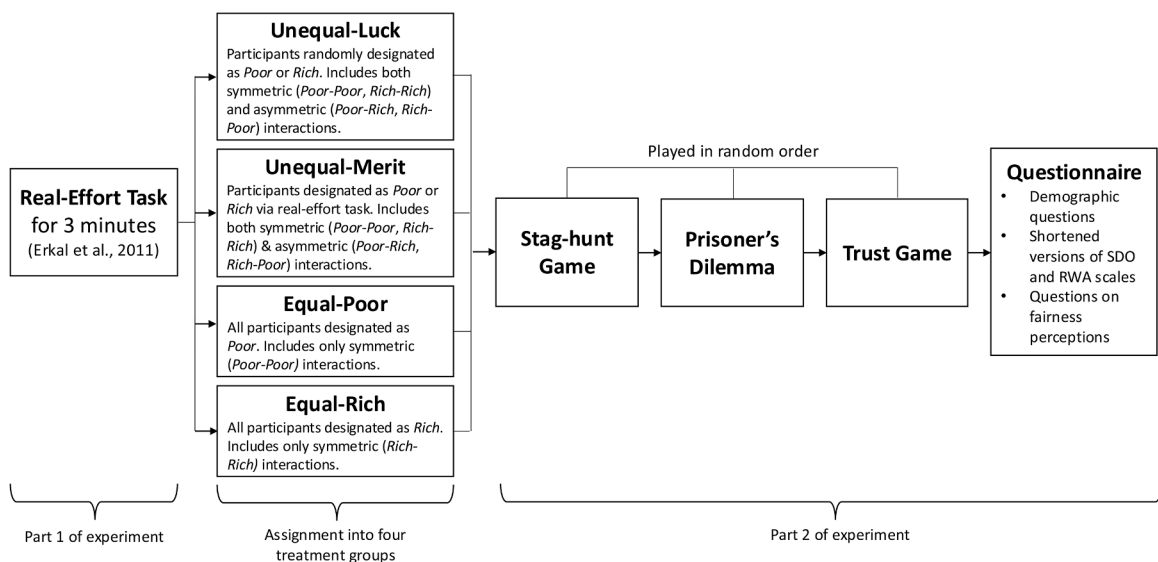
#### 3.1. Real effort task

The experiment proceeded in two parts. In Part 1, participants engaged in a real effort “encryption task” requiring cognitive effort (Erkal et al., 2011). Participants had three minutes to convert as many random strings of three letters as possible to numeric codes using an “encryption table” with a set of unique numbers corresponding to each letter of the alphabet. Participants received a piece rate of 5 tokens per correct code submitted, with no deductions for errors. This piece rate applied to all participants, irrespective of treatment, and they kept these earnings in addition to any payoffs from the subsequent games. To avoid potential wealth effects, participants were informed of their real-effort task score and earnings from Part 1 only at the end of the entire experiment. While participants knew the study involved two parts, they received instructions for Part 1 alone, with no indication of what Part 2 would entail or whether their performance or earnings would matter in later parts. Appendix C contains all the instructions given to participants.

#### 3.2. Treatments

Following the real-effort task, participants were randomly assigned to one of four treatment groups using a between-subjects design: *Equal-Poor*, *Equal-Rich*, *Unequal-Luck* and *Unequal-Merit*. Each participant received an endowment of either 50 tokens (*poor*) or 100 tokens (*rich*), a status that remained fixed throughout the experiment. Treatments differed in whether there was inequality between participants and, if so, how such inequality was generated.

In the equal treatments, all participants received uniform token endowments, either 50 or 100 tokens. In *Equal-Poor* (*Equal-Rich*), each participant was endowed with 50 tokens (100 tokens). In *Unequal-Luck*, half the participants were randomly chosen to be endowed with 50 tokens, while the other half were assigned 100 tokens, regardless of effort (or earnings) in the real-effort task. In *Unequal-Merit*, endowments were linked to performance in the real-effort task performed in Part 1. Participants who scored above 19 (i.



**Fig. 1.** Overview of the experimental design showing assignment to treatment groups. Part 1 consists of the real effort task. Part 2 includes the three games presented in random order. A questionnaire is administered at the end of the session.



e., the median score obtained from pilot data), received 100 tokens, whereas those scoring equal or below this threshold were allocated 50 tokens. In both the *Equal* treatments and the *Unequal-Luck* treatment, performance in the real-effort task had no bearing on subsequent endowments. However, unlike the *Unequal* treatments, participants in the *Equal* treatments knew they would receive identical endowments and interact solely with others of the same status. Thus, they were not exposed to asymmetric interactions, in contrast to their counterparts in the *Unequal* treatments.

Participants remained in their assigned treatment group throughout the experiment. While we informed them about their treatment's endowment allocation process, we did not disclose the existence of other treatment groups. For payments, tokens were converted at a rate of 250 tokens to £1. To determine individual earnings, we formed participant pairs ex post by using the strategy method (Brandts and Charness, 2000; Mitzkewitz and Nagel, 1993). Accordingly, participants submitted their responses to each game at the outset, which we later matched with the decisions of others within their respective treatment. See Fig. 1 for an overview of the experimental design.

### 3.3. Economic games

Once participants were assigned to the four treatments, they proceeded to Part 2 of the experiment, where they played the trust game, the stag-hunt game and the prisoner's dilemma. The games were presented in random order to minimise order effects. In the *Equal-Rich* treatment, *rich* participants (endowed with 100 tokens) engaged in the games paired with other *rich* participants (i.e., *rich-rich* pairs), and similarly, in the *Equal-Poor* treatment, *poor* participants (endowed with 50 tokens) were paired with other *poor* participants (*poor-poor* pairs). Thus, in both equal treatments, endowments were homogeneous, allowing only symmetric pairings. In contrast, the two unequal treatments included both symmetric and asymmetric pairings. Participants in the *Unequal-Merit* and *Unequal-Luck* treatments played each game twice, once when paired with a peer of the same endowment (i.e., *rich* interacting with *rich* and *poor* with *poor*) and once with a peer of a different endowment level (i.e., *rich* interacting with *poor* and vice versa). As a result, participants in the unequal treatments made twice as many decisions as those in the equal treatments, with the sequence of symmetric and asymmetric pairings randomised. Crucially, in all cross-treatment comparisons, we hold pairings (and therefore endowments) constant, allowing us to isolate the treatment effect from potential confounds.

#### 3.3.1. Trust game

This is a two-player sequential move game. The first mover could transfer a share of their endowment to the second mover. The experimenter tripled this transferred amount. The second mover could return any share of this tripled amount back to the first mover. The returned amount is not tripled. The game ends after the second mover's decision. In our study, each participant acted as both the first and the second mover. As first movers, they decided how much of their endowment to send to the second mover in increments of 10 per cent of their total endowment (e.g., 0, 5, 10, 15, ... etc. if the endowment was 50; or 0, 10, 20, 30, ... etc. if the endowment was 100). This amount served as a proxy for the first movers' trust.

As second movers, participants indicated how much they would return, conditional on each possible tripled amount they could receive from the first mover. From these responses, we calculated each participant's average return rate. This average proportion served as an individual-level measure of trustworthiness. Assuming purely self-regarding preferences, the subgame perfect Nash equilibrium, via backward induction, is for the second mover to return zero, and, anticipating this, for the first mover to send zero.

Before making decisions, participants completed comprehension checks to ensure they understood the task. Incorrect responses triggered automated explanations, and participants reattempted the question until they answered all questions correctly. Only then could they proceed to the decision stage. Note that in the equal treatment, participants made two decisions: once as the first mover and once as the second mover. In the Unequal treatments, they made four decisions: two as first mover (once with a partner of the same endowment and once with a partner of a different endowment) and two as second mover under the same pairing conditions. The order of these decisions was randomised across participants to minimise potential order effects.

#### 3.3.2. Stag-hunt game and the prisoner's dilemma

In the stag-hunt game, participants simultaneously chose to either 'invest' 60 per cent of their endowment or 'keep' their entire endowment. If both invested, each player earned their own endowment plus the combined investments of both player. If neither invested, both simply retained their initial endowment. If only one invested, the investor lost this investment while the non-investor gained this amount.

Table 1 summarises these rules in a general payoff matrix that applies to all possible pairings between *rich* (endowment,  $y = 100$ ) and *poor* (endowment,  $y = 50$ ) participants. The game has two Nash equilibria: {Invest, Invest}, which is payoff-dominant, and {Keep, Keep}, which is secure. The resulting payoff matrices for all possible pairings are shown in Appendix B, Table B1.

In the prisoner's dilemma, participants also began with an endowment of either 50 or 100 tokens. Each could choose to 'take', which meant deducting 60 per cent of their partner's endowment. The player who took received half of the deducted amount, while the partner bore the full loss. If both took, each lost 60 per cent of their own endowment but simultaneously gained half of 60 per cent of their partner's endowment. If neither took, both simply retained their endowment.

Table 2 shows the general payoff matrix for all possible pairings between *rich* and *poor* participants. In all cases, this game has a unique dominant strategy equilibrium of {Take, Take}. The four payoff matrices resulting from this for each possible pairing type are shown in Appendix B, Table B2. As in the trust game, participants completed comprehension checks before proceeding to their decisions. Explanations were provided after any incorrect responses, and participants advanced only after answering all questions correctly.

**Table 1**

Payoffs for all possible pairings in the stag hunt game for participants with endowment  $y_i, y_j \in \{50, 100\}$ . Mutual investment yields the payoff-dominant outcome.

|                                       |        | Participant j<br>[Endowment = $y_j$ ]            |                                |
|---------------------------------------|--------|--------------------------------------------------|--------------------------------|
|                                       |        | Invest                                           | Keep                           |
| Participant i<br>[Endowment = $y_i$ ] | Invest | $(y_i + 0.6y_i + 0.6y_j, y_j + 0.6y_i + 0.6y_j)$ | $(y_i - 0.6y_i, y_j + 0.6y_i)$ |
|                                       | Keep   | $(y_i + 0.6y_j, y_j - 0.6y_j)$                   | $(y_i, y_j)$                   |

Note: The separate payoff matrices for each specific *rich/poor* pairing are provided in [Appendix B, Table B1](#).

**Table 2**

Payoffs for all possible pairings in the prisoner's dilemma game participants with endowment  $y_i, y_j \in \{50, 100\}$ . Mutual taking yields the Nash equilibrium.

|                                       |          | Participant j<br>[Endowment = $y_j$ ]                                        |                                              |
|---------------------------------------|----------|------------------------------------------------------------------------------|----------------------------------------------|
|                                       |          | Take                                                                         | Not take                                     |
| Participant i<br>[Endowment = $y_i$ ] | Take     | $(y_i - 0.6y_i + \frac{1}{2} * 0.6y_j, y_j - 0.6y_j + \frac{1}{2} * 0.6y_i)$ | $(y_i + \frac{1}{2} * 0.6y_j, y_j - 0.6y_j)$ |
|                                       | Not take | $(y_i - 0.6y_i, y_j + \frac{1}{2} * 0.6y_i)$                                 | $(y_i, y_j)$                                 |

Note: The separate payoff matrices for each specific *rich/poor* pairing are provided in [Appendix B, Table B2](#).

### 3.3.3. Belief elicitation

We also elicited participants' beliefs about their counterparts' actions. In the trust game, we asked first movers how many tokens they believed the second mover would send back to them (based on knowing what they had decided to send) as well as their estimate of the average amount sent by other first movers in similar conditions. To all second movers, we asked how many tokens they believed the majority of other second movers returned upon receiving 90 tokens. Participants received an additional 10 tokens as a reward if their predictions fell within a 5-token margin of the actual observed behaviour, aligning with [Gächter and Renner's \(2010\)](#) methodology for belief elicitation. In the stag-hunt and prisoner's dilemma games, participants estimated the majority's likely actions: 'invest' or 'keep', 'take' or 'not take'. Correct predictions were similarly incentivised with a 10-token reward. For parsimony, we omit the analysis of belief data. While beliefs across games are of independent interest, they did not yield clear insights into how treatments shaped expectations of others' play. As our focus is on treatment effects, including these results would add little to the main analysis. All related analyses are available upon request.

### 3.3.4. Final questionnaire

After submitting their choices across all three games, the participants filled out a questionnaire which comprised five demographic questions (age, gender, country of birth, education and employment status) and shortened versions of the Social Dominance Orientation (SDO) ([Pratto et al., 1994](#)) and the Right-Wing Authoritarianism (RWA) scale ([Altemeyer, 1981](#)). High SDO scores indicate greater tolerance of intergroup inequality while higher RWA scores indicate greater social conformity.

In the questionnaire, we also assessed participants' fairness perceptions of the three endowment allocation methods: equal, luck-based and merit-based. We first asked them to rate the fairness of the endowment allocation method used in their treatment group on a five-point Likert scale with 1 indicating "very unfair" to 5 indicating "very fair." Next, we disclosed the other endowment allocation methods used in the study and then asked them to re-evaluate the fairness of their endowment allocation method using the same question. This repeated assessment aims to capture the influence of awareness about alternative allocation methods on participants' fairness judgments. Finally, we asked participant to choose what they considered the fairest allocation method except they were given four options, namely, equal, luck-based, merit-based and an uncertain method. The uncertain endowment allocation method in this context simply means a 50 per cent chance of the allocation being determined based on luck and a 50 per cent chance of it being determined by merit. We added this fourth option primarily as a way to avoid the compromise effect, a bias towards the middle option ([Beauchamp et al., 2020](#)).

## 3.4. Participants

We initially recruited 1212 adult participants through Prolific, a UK-based online crowdsourcing platform, across eight sessions. Of these, 168 individuals started the study but dropped out before completing, resulting in a final sample of 1044 participants. A detailed breakdown of participant attrition by treatment group and experimental stage is provided in [Table B3](#) in [Appendix B](#). Our only inclusion criterion was that the participants must be aged above 18 years. Approximately 54 per cent identified as female and the average age is 26.93 years. A total of 34 countries were listed as participants' country of birth.<sup>3</sup> The experiment took approximately 25 min to

<sup>3</sup> In [Appendix A](#), we present balance tables on key demographic variables.

complete. Participants earned an average of £4.17, which consisted of a fixed £2.50 participation fee in addition to the variable payments which were based on their treatment, own decisions and partner decisions. This corresponds to an hourly rate of £10 or US \$13. Prolific insists that participants be paid a minimum of £6/hour. To adhere to this rule, we paid a fixed £2.50 participation fee for the 25-minute task.

Table 3 gives the breakdown of the number of participants and average earnings based on the treatment groups.<sup>4</sup> The experiment was approved by the University of Auckland Human Participants Ethics Committee (Ref. No.: UAHPEC23602). Raw data for this study can be accessed at [data.mendeley.com/datasets/8gsjdwtxbx/1](https://data.mendeley.com/datasets/8gsjdwtxbx/1).

### 3.5. Power calculations

We performed power calculations using 42 observations from a pilot study to establish the sample size needed to detect a significant difference between luck-based and merit-based inequality. The pilot comprised 19 observations from the *Unequal-Merit* and 23 from the *Unequal-Luck* group.<sup>5</sup> We did not modify the experimental design after the pilot, and as a result, the pilot session followed the same design as outlined above. We aimed for a sample size that would provide 80 per cent power at a 5 per cent significance level ( $\alpha$ ), minimising the risk of type II errors in measuring social capital across three games. Recall from our hypotheses, that we expected higher trust levels in the trust game, a greater percentage of ‘invest’ decisions in the stag-hunt game, and fewer ‘take’ decisions in the prisoner’s dilemma under merit. Given the limited size of our pilot data, we employed Monte Carlo simulations for a robust analysis. We evaluated the treatments as a whole, disregarding participant endowment. To detect a merit effect with 80 per cent power based on the estimated effects in the pilot, our study needed 200 participants per treatment in the prisoner’s dilemma, 100 in the stag-hunt game, and 350 in the trust game. Consequently, we aimed for 350 participants per treatment. We ended up with 352 participants in the equal (including *Equal-Rich* and *Equal-Poor*) and *Unequal-Luck* treatments each and 340 participants in the *Unequal-Merit* treatment.

## 4. Results

In Section 4.1, we test our first hypothesis by considering whether inequality, irrespective of its source, reduces social capital across the three games. Then, in Section 4.2, we examine our second hypothesis, i.e., whether luck-based inequality results in a greater decline in social capital compared to merit. Finally, in Section 4.3, we analyse participants’ fairness evaluations of the different allocation mechanisms. Summary statistics and results from the balance of treatment tests are presented in Appendix A.

### 4.1. Support for Hypothesis 1: impact of inequality irrespective of its source

In this section, we examine behavioural differences between participants in equal and unequal treatments, regardless of the source of inequality. Accordingly, we combined the decisions of participants from both the *Unequal-Luck* and *Unequal-Merit* treatments, referring to them collectively as “Unequal” societies, and compare these to decisions made in the *Equal-Poor* and *Equal-Rich* treatments, which we collectively label as “Equal” societies. To isolate the effect of inequality on social capital and avoid potential confounds, we restrict our analysis to interactions between participants with identical endowments (i.e., *rich* interacting with *rich* and *poor* with *poor*). By holding pairings constant, we eliminate confounding effects of individual wealth asymmetries, allowing us to attribute any observed behavioural differences to the influence of societal inequality alone.

In Table 4, we present non-parametric test results designed to test support (or lack thereof) for Hypotheses 1. Hypothesis 1 predicts that inequality erodes social capital relative to equality, holding payoff structures constant. This hypothesis yields four testable outcomes: inequality should (i) lower trust in the trust game, (ii) lower trustworthiness in the trust game, (iii) lower rates of investment in the stag-hunt game, and (iv) increase rates of taking in the prisoner’s dilemma game. We present pooled results across the two types of societies as well as disaggregated results broken up by endowment status (i.e., for *poor-poor* and *rich-rich* pairings in each society). Since trust and trustworthiness are bounded continuous variables, we use Wilcoxon rank sum tests to see if the observations for equal and unequal societies are drawn from different distributions. Investment in the stag hunt game and taking in the prisoner’s dilemma are binary decisions, and therefore we use Chi-squared tests to compare proportions across treatments.

As we evaluate Hypothesis 1 across four distinct outcomes (i.e., trust, trustworthiness, investment and taking rates), we correct for multiple hypothesis testing. Specifically, we apply a Bonferroni (Bonf) correction to control for the family-wise error rate (FWER) as well as a Benjamini-Hochberg (BH) correction to control for the false discovery rate (FDR).

Based on Table 4, and focusing on the pooled comparisons, inequality is associated with significantly lower trustworthiness and lower investment rate in the stag-hunt game (the latter implying lower coordination on the payoff dominant equilibrium). In particular, trustworthiness declines in unequal societies (pooled raw p-value = 0.009; Bonf-adjusted p-value = 0.037; BH-adjusted p-

<sup>4</sup> We ended up with an odd number of *rich* and *poor* participants in a few cases preventing the formation of unique symmetric pairings. To resolve this, we paired one participant in each treatment with two other randomly picked participants with the same endowment for final earnings calculations. This did not affect results, as decisions are elicited using the strategy method and all pairings were made ex post.

<sup>5</sup> In the pilot study, the *Unequal* treatments had an imbalance between *rich* and *poor* participants. To address this, some participants were paired with several randomly chosen others to determine final earnings. This did not affect results, as decisions are elicited using the strategy method and all pairings were made ex post.



**Table 3**

Summary of treatments showing the number of participants and average earnings in each sub-group.

| Treatment     | Endowment before each game | Status | Number of participants | Average earnings |
|---------------|----------------------------|--------|------------------------|------------------|
| Equal         | 50 tokens                  | Poor   | 163                    | £3.38            |
| Equal         | 100 tokens                 | Rich   | 189                    | £4.13            |
| Unequal-Merit | 50 tokens                  | Poor   | 182                    | £3.25            |
|               | 100 tokens                 | Rich   | 158                    | £3.98            |
| Unequal-Luck  | 50 tokens                  | Poor   | 165                    | £3.4             |
|               | 100 tokens                 | Rich   | 187                    | £3.87            |

**Table 4**

Comparing outcomes between the Equal and Unequal societies.

| Outcome                                                                               | Pairings | Equal Societies      | Unequal Societies    | p-value (raw)       | p-value (Bonf) | p-value (BH) |
|---------------------------------------------------------------------------------------|----------|----------------------|----------------------|---------------------|----------------|--------------|
| Trust<br>Percent of endowment sent by first mover in the trust game                   | All      | 37.24 %<br>(n = 352) | 36.91 %<br>(n = 692) | 0.833 <sup>a</sup>  | 1.0            | 0.833        |
|                                                                                       | Poor     | 43.37 %<br>(n = 163) | 42.88 %<br>(n = 347) | 0.978 <sup>a</sup>  | 1.0            | 1.0          |
|                                                                                       | Rich     | 31.96 %<br>(n = 189) | 30.90 %<br>(n = 345) | 0.322 <sup>a</sup>  | 1.0            | 1.0          |
| Trustworthiness<br>Percent returned by second mover in the trust game                 | All      | 44.01 %<br>(n = 352) | 41.54 %<br>(n = 692) | 0.009 <sup>a</sup>  | 0.037          | 0.018        |
|                                                                                       | Poor     | 43.88 %<br>(n = 163) | 40.92 %<br>(n = 347) | 0.040 <sup>a</sup>  | 0.16           | 0.08         |
|                                                                                       | Rich     | 44.11 %<br>(n = 189) | 42.17 %<br>(n = 345) | 0.099 <sup>a</sup>  | 0.396          | 1.0          |
| Investment Rate<br>Percent of participants choosing “invest” in the stag hunt game    | All      | 82.39 %<br>(n = 352) | 72.98 %<br>(n = 692) | <0.001 <sup>b</sup> | 0.004          | 0.004        |
|                                                                                       | Poor     | 83.44 %<br>(n = 163) | 71.76 %<br>(n = 347) | 0.006 <sup>b</sup>  | 0.024          | 0.024        |
|                                                                                       | Rich     | 81.48 %<br>(n = 189) | 74.20 %<br>(n = 345) | 0.072 <sup>b</sup>  | 0.284          | 1.0          |
| Taking Rate<br>Percent of participants choosing “take” in the prisoner’s dilemma game | All      | 56.53 %<br>(n = 352) | 55.49 %<br>(n = 692) | 0.799 <sup>b</sup>  | 1.0            | 1.0          |
|                                                                                       | Poor     | 53.99 %<br>(n = 163) | 53.60 %<br>(n = 347) | 1.0 <sup>b</sup>    | 1.0            | 1.0          |
|                                                                                       | Rich     | 58.73 %<br>(n = 189) | 57.39 %<br>(n = 345) | 0.835 <sup>b</sup>  | 1.0            | 1.0          |

Notes: Cells report mean percentages; sample sizes appear in parentheses. “Equal societies” refer to the combined Equal-Poor and Equal-Rich treatments. “Unequal societies” pool data from the Merit- and Luck-based unequal treatments. To isolate the effect of group-level inequality without endowment and payoff confounds, we compare only symmetric interactions (i.e., *rich* interacting with *rich* and *poor* with *poor*) within both unequal and equal groups.

<sup>a</sup> Wilcoxon rank sum test.

<sup>b</sup> Chi-squared test. Bonf denotes Bonferroni correction (family-wise error rate control). BH denotes Benjamini-Hochberg correction (false discovery rate control). Corrected p-values are truncated at 1.0 when the adjusted value exceeds 1.

value = 0.018). Similarly, coordination on the payoff-dominant equilibrium in the stag-hunt game is significantly lower under inequality (pooled raw p-value = 0.0009; Bonf-adjusted p-value = 0.004; BH-adjusted p-value = 0.004). These pooled effects are driven primarily by *poor-poor* interactions, although the overall direction of effects is consistent across endowment groups.

Since non-parametric tests do not control for other covariates, in Table 5 we report parametric regression results for the four outcomes of interest. For trust and trustworthiness, which are continuous variables, we report ordinary least square (OLS) estimates. For investment in the stag-hunt and taking in the prisoner’s dilemma, which are binary choices, we run probit models. We ran two specifications for each outcome. The first specification incorporates a dummy for inequality and a dummy for *rich* status. In the second specification, we incorporate a comprehensive set of controls including the encoding score in the real-effort task, demographics, prior experience with economic games, scores for SDO and RWA, and the order in which participants played the three games (to check for potential order effects).

Consistent with the non-parametric results reported earlier, the regression analysis confirms that inequality has a negative effect on trustworthiness. In the first specification, the coefficient on the Unequal dummy is negative and statistically significant at conventional levels. In the second specification, which includes the full set of controls, the coefficient remains negative and economically similar in magnitude, although its statistical significance weakens (p-value = 0.089). Importantly, the direction of the effect is stable across the two specifications. In terms of outcomes in the stag-hunt game, the probability of choosing to invest in the stag-hunt game remains significantly lower in unequal societies across both specifications. While not directly relevant to this hypothesis, we also note that in general, the *rich* are less trusting but there are no significant differences in the other outcomes between the *rich* and *poor*.

We also correct for multiple hypothesis testing across the four main outcome variables within each regression specification. The

**Table 5**

Ordinary least square (OLS) regression results for trust game decisions (trust and trustworthiness), and probit regression results for investment in the stag-hunt game and taking in the prisoner's dilemma.

|                                                                         | Trust                 |                       | Trustworthiness      |                      | Investment Rate<br>Invest = 1;<br>Keep = 0 |                      | Taking Rate<br>Take = 1;<br>Not take = 0 |                      |
|-------------------------------------------------------------------------|-----------------------|-----------------------|----------------------|----------------------|--------------------------------------------|----------------------|------------------------------------------|----------------------|
|                                                                         | (1)                   | (2)                   | (1)                  | (2)                  | (1)                                        | (2)                  | (1)                                      | (2)                  |
| Intercept                                                               | 43.576***<br>(1.69)   | 57.775***<br>(11.035) | 43.519***<br>(1.153) | 33.846***<br>(7.627) | 0.914***<br>(0.091)                        | 0.991*<br>(0.582)    | 0.109<br>(0.079)                         | 0.331<br>(0.536)     |
| Unequal treatment                                                       | -0.789<br>(1.763)     | 0.14<br>(1.728)       | -2.427**<br>(1.202)  | -2.033*<br>(1.194)   | -0.317***<br>(0.094)                       | -0.322***<br>(0.096) | -0.023<br>(0.083)                        | -0.06<br>(0.084)     |
| Rich status                                                             | -11.793***<br>(1.667) | -12.803***<br>(1.682) | 0.906<br>(1.137)     | -0.014<br>(1.163)    | 0.03<br>(0.086)                            | -0.018<br>(0.091)    | 0.104<br>(0.078)                         | 0.155*<br>(0.082)    |
| Encoding Score                                                          |                       | 0.233<br>(0.144)      |                      | 0.252**<br>(0.099)   |                                            | 0.005<br>(0.008)     |                                          | -0.022***<br>(0.007) |
| Controls: demographics, prior<br>experimental experience, SDO, &<br>RWA | No                    | Yes                   | No                   | Yes                  | No                                         | Yes                  | No                                       | Yes                  |
| Control for Order Effects                                               | No                    | Yes                   | No                   | Yes                  | No                                         | Yes                  | No                                       | Yes                  |
| Obs. <sup>a</sup>                                                       | 1044                  | 1044                  | 1044                 | 1044                 | 1044                                       | 1044                 | 1044                                     | 1044                 |
| R <sup>2</sup> / R <sup>2</sup> adjusted                                | 0.046 /<br>0.044      | 0.117 /<br>0.097      | 0.005 /<br>0.003     | 0.054 /<br>0.032     | -                                          | -                    | -                                        | -                    |
| Nagelkerke R <sup>2</sup>                                               | -                     | -                     | -                    | -                    | 0.017                                      | 0.077                | 0.002                                    | 0.056                |
| AIC                                                                     | -                     | -                     | -                    | -                    | 1141.13                                    | 1141.92              | 1437.11                                  | 1438.60              |

Notes: For each outcome, specification (1) reports a baseline specification including the Unequal treatment dummy and Rich status dummy. Specification (2) adds encoding score, demographic controls (age, gender, education, employment), prior experimental experience, SDO and RWA scores, and order controls.

Trust and trustworthiness are estimated using OLS. Investment in the stag-hunt game (invest = 1; keep = 0) and taking in the prisoner's dilemma (take = 1; not take = 0) are estimated using probit models.

"Unequal treatment" equals 1 for participants in the Merit or Luck treatments and 0 for those in the Equal treatments. "Rich status" equals 1 for participants endowed with 100 tokens and 0 for those endowed with 50. Figures in parentheses are standard errors. Each observation represents an independent decision from participants paired symmetrically in the equal and unequal treatments.

Significance levels:

\*\*\* p < 0.01.

\*\* p < 0.05.

\* p < 0.1.

corrected p-values are reported in [Appendix B, Table B5](#). The negative effect of inequality on investment rate in the stag-hunt game remains statistically significant under both correction methods in both specifications (Bonf-adjusted p-values = 0.003 in both models; BH-adjusted p-values = 0.003 in both models). In contrast, the effect on trustworthiness, while statistically significant in the baseline specification (raw p-value = 0.044), becomes weakly significant after adjustment (Bonf-adjusted p-value = 0.175; BH-adjusted p-value = 0.087) and becomes insignificant in the fully controlled specification.

Overall, we find partial support for Hypothesis 1. Our results suggest that inequality leads to lower levels of trustworthiness in the trust game and lower coordination to the payoff dominant equilibrium in the stag hunt game. The former result, related to trustworthiness, is weaker, as it is more sensitive to controls and to multiple hypotheses testing corrections. Taken together, these findings suggest that inequality undermines some aspects of social capital, particularly coordination and trustworthiness.

#### 4.2. Support for Hypothesis 2: role of the source of inequality

In this section, we focus on how the source of inequality shapes behaviour. To do so, we restrict our analysis to the 692 participants from the *Unequal-Merit* and *Unequal-Luck* treatments, drawn from the initial sample of 1044.

Our experimental design varies two factors: endowment status (*rich* or *poor*) and the source of inequality (merit or luck). To isolate the effect of the source, we first present results from non-parametric tests comparing only the symmetric pairings (i.e., *rich-rich* and *poor-poor*) between the merit and luck treatments. As before, we rely on the Wilcoxon rank sum test when comparing trust and trustworthiness between the two sources of inequality, luck and merit. For rates of investment in the stag-hunt game and taking in the prisoner's dilemma game, we use Chi-squared tests. These results are reported in [Table 6](#).

**Table 6**Comparing outcomes between merit-based and luck-based inequality for only the symmetric pairings (*rich-rich* and *poor-poor* pairings).

| Outcome                                                                               | Pairings | Merit                | Luck                 | p-value (raw)      | p-value (Bonf) | p-value (BH) |
|---------------------------------------------------------------------------------------|----------|----------------------|----------------------|--------------------|----------------|--------------|
| Trust<br>Percent of endowment sent by first mover in the trust game                   | All      | 34.74 %<br>(n = 340) | 39.01 %<br>(n = 352) | 0.143 <sup>a</sup> | 0.572          | 0.286        |
|                                                                                       | Poor     | 40.55 %<br>(n = 182) | 45.45 %<br>(n = 165) | 0.129 <sup>a</sup> | 0.515          | 0.257        |
|                                                                                       | Rich     | 28.04 %<br>(n = 158) | 33.32 %<br>(n = 187) | 0.162 <sup>a</sup> | 0.647          | 0.647        |
| Trustworthiness<br>Percent returned by second mover in the trust game                 | All      | 41.46 %<br>(n = 340) | 41.62 %<br>(n = 352) | 0.782 <sup>a</sup> | 1.0            | 0.782        |
|                                                                                       | Poor     | 40.18 %<br>(n = 182) | 41.74 %<br>(n = 165) | 0.630 <sup>a</sup> | 1.0            | 0.630        |
|                                                                                       | Rich     | 42.94 %<br>(n = 158) | 41.52 %<br>(n = 187) | 0.380 <sup>a</sup> | 1.0            | 0.761        |
| Investment Rate<br>Percent of participants choosing “invest” in the stag hunt game    | All      | 71.18 %<br>(n = 340) | 74.72 %<br>(n = 352) | 0.336 <sup>b</sup> | 1.0            | 0.448        |
|                                                                                       | Poor     | 69.23 %<br>(n = 182) | 74.55 %<br>(n = 165) | 0.328 <sup>b</sup> | 1.0            | 0.437        |
|                                                                                       | Rich     | 73.42 %<br>(n = 158) | 74.87 %<br>(n = 187) | 0.855 <sup>b</sup> | 1.0            | 0.855        |
| Taking Rate<br>Percent of participants choosing “take” in the prisoner’s dilemma game | All      | 52.35 %<br>(n = 340) | 58.52 %<br>(n = 352) | 0.120 <sup>b</sup> | 0.479          | 0.285        |
|                                                                                       | Poor     | 48.90 %<br>(n = 182) | 58.79 %<br>(n = 165) | 0.082 <sup>b</sup> | 0.330          | 0.257        |
|                                                                                       | Rich     | 56.33 %<br>(n = 158) | 58.29 %<br>(n = 187) | 0.797 <sup>b</sup> | 1.0            | 0.855        |

Notes: Cells report mean percentages; sample sizes appear in parentheses. “Symmetric pairings” refer to interactions between participants with identical endowments (*poor-poor* and *rich-rich*).

<sup>a</sup> Wilcoxon rank sum test. <sup>b</sup> Chi-squared test.

Bonf denotes Bonferroni correction (family-wise error rate control). BH denotes Benjamini-Hochberg correction (false discovery rate control). Corrected p-values are truncated at 1.0 when the adjusted value exceeds 1.

We then extend the analysis in Table 7 to asymmetric pairings, namely *rich-poor* and *poor-rich* interactions, across the merit and luck treatments. Together, Tables 6 and 7 allow us to examine all four possible pairing types, *rich-rich*, *poor-poor*, *rich-poor*, and *poor-rich*, under both sources of inequality while holding the pairing structure fixed in each comparison. This strategy guarantees that we are changing one factor at a time, allowing for a clean test of how the source of inequality influences social capital.<sup>6</sup>

Contrary to our hypothesis, we observe higher trust across all participant pairings under luck compared to merit, but the differences are not significant at the 5 per cent level. Only the pooled asymmetric group (*rich-poor* and *poor-rich* combined) and the *rich* asymmetric group (*rich-poor*) show significance at the 10 per cent level in the unadjusted tests. The source of inequality does not significantly impact trustworthiness, investment in the stag hunt game or taking in the prisoner’s dilemma game.

Non-parametric tests provide useful descriptive insights but have clear limitations: they do not control for covariates or repeated decisions made by the same participant. To address this, we next present parametric regression results for all four outcomes of interest using a mixed-effects framework. As before, we analyse trust, trustworthiness, investment in the stag-hunt game, and taking in the prisoner’s dilemma within a unified regression structure. The key difference compared to the previous section is that we now employ linear mixed-effects models for the continuous outcomes (trust and trustworthiness) and mixed-effects probit models for the binary outcomes (investment and taking).

In all specifications, the baseline is *poor* participants in the merit treatment. We incorporated participant ID as a random effect to account for the interdependence of repeated observations from the same participant under symmetric and asymmetric pairings. This adjustment ensures that within-subject variations do not bias our estimates. As before, we report two specifications for each outcome. The first includes a dummy indicator for the luck treatment and a dummy for rich status. The second specification adds a comprehensive set of controls, including performance in the real-effort task, demographic characteristics, prior experience with economic games, SDO and RWA scores, and order effects. The results are presented in Table 8.<sup>7</sup>

<sup>6</sup> The comparison between symmetric and asymmetric pairings is certainly of interest, but we will leave that analysis for separate work as it is beyond the scope of this present paper.

<sup>7</sup> It is worth noting that when we conduct OLS regressions on split datasets without repeated observations (i.e., one with choices from symmetric interactions and the other using asymmetric interactions) we observe qualitatively consistent results.

**Table 7**Comparing under merit-based and luck-based inequality for only the asymmetric pairings (*rich-poor* and *poor-rich* pairings).

| Outcome                                                                               | Pairings | Merit                | Luck                 | p-value (raw)      | p-value (Bonf) | p-value (BH) |
|---------------------------------------------------------------------------------------|----------|----------------------|----------------------|--------------------|----------------|--------------|
| Trust<br>Percent of endowment sent by first mover in the trust game                   | All      | 32.79 %<br>(n = 340) | 37.10 %<br>(n = 352) | 0.085 <sup>a</sup> | 0.338          | 0.338        |
|                                                                                       | Poor     | 37.69 %<br>(n = 182) | 42.30 %<br>(n = 165) | 0.152 <sup>a</sup> | 0.609          | 0.609        |
|                                                                                       | Rich     | 27.15 %<br>(n = 158) | 32.51 %<br>(n = 187) | 0.072 <sup>a</sup> | 0.287          | 0.287        |
| Trustworthiness<br>Percent returned by second mover in the trust game                 | All      | 40.42 %<br>(n = 340) | 41.56 %<br>(n = 352) | 0.499 <sup>a</sup> | 1.0            | 0.904        |
|                                                                                       | Poor     | 37.82 %<br>(n = 182) | 38.94 %<br>(n = 165) | 0.634 <sup>a</sup> | 1.0            | 0.846        |
|                                                                                       | Rich     | 43.41 %<br>(n = 158) | 43.88 %<br>(n = 187) | 0.817 <sup>a</sup> | 1.0            | 0.896        |
| Investment Rate<br>Percent of participants choosing “invest” in the stag hunt game    | All      | 61.47 %<br>(n = 340) | 61.65 %<br>(n = 352) | 1.0 <sup>b</sup>   | 1.0            | 1.0          |
|                                                                                       | Poor     | 62.09 %<br>(n = 182) | 61.21 %<br>(n = 165) | 0.955 <sup>b</sup> | 1.0            | 0.955        |
|                                                                                       | Rich     | 60.76 %<br>(n = 158) | 62.03 %<br>(n = 187) | 0.896 <sup>b</sup> | 1.0            | 0.896        |
| Taking Rate<br>Percent of participants choosing “take” in the prisoner’s dilemma game | All      | 68.53 %<br>(n = 340) | 66.76 %<br>(n = 352) | 0.678 <sup>b</sup> | 1.0            | 0.904        |
|                                                                                       | Poor     | 73.08 %<br>(n = 182) | 77.58 %<br>(n = 165) | 0.398 <sup>b</sup> | 1.0            | 0.796        |
|                                                                                       | Rich     | 63.29 %<br>(n = 158) | 57.22 %<br>(n = 187) | 0.300 <sup>b</sup> | 1.0            | 0.600        |

Notes: Cells report mean percentages; sample sizes appear in parentheses. “Asymmetric pairings” refer to interactions between participants with different endowments (*poor-rich* and *rich-poor*).

<sup>a</sup> Wilcoxon rank sum test.

<sup>b</sup> Chi-squared test.

Bonf denotes Bonferroni correction (family-wise error rate control). BH denotes Benjamini-Hochberg correction (false discovery rate control). Corrected p-values are truncated at 1.0 when the adjusted value exceeds 1.

We find that contrary to our hypothesis that trust should be higher in merit-based societies, it is actually the luck treatment that positively influences trust in all specifications. For all the other outcomes (trustworthiness, investment rate and taking rate), we find no treatment effect. In addition, we adjust for multiple hypothesis testing across the four primary outcome variables within each regression specification. The corrected p-values are reported in [Appendix B, Table B6](#). We use the same two model specifications as before (i.e., one with a treatment dummy and an endowment dummy, and a second one that included the full set of controls). After applying the Bonferroni and the Benjamini-Hochberg corrections, the effect remains statistically significant in the baseline specification (adjusted  $p = 0.028$ ) and is marginally significant in the fully controlled specification (adjusted  $p = 0.086$ ).

To further explore this unexpected result that trust is higher in the luck treatment, we conducted OLS regressions on subsamples categorised by pairing types (*poor-poor*, *poor-rich*, *rich-rich* and *rich-poor*). The results, shown in [Table 9](#), reveal that the *rich* consistently display greater trust in the luck treatment, regardless of whether they are paired with other *rich* or *poor* participants.<sup>8</sup>

#### 4.2.1. Why is trust higher under luck than merit?

One potential interpretation of the higher trust observed under luck-based inequality than under merit-based inequality is that it reflects redistributive motives. If this were the case, then we would expect fairness perceptions to predict transfers. Transfers should be relatively high when an allocation is considered unfair while transfers should be relatively low when an allocation is considered fair. In other words, first mover transfers should decline as fairness perception scores changes from 1 (very unfair) to 5 (very fair) and this should apply particularly to transfers from *rich* first movers to *poor* second movers.

The evidence is mixed. In the luck treatment, fairness perceptions show no systematic link to trust ([Fig. B1, Appendix B](#)), and mixed-effects regressions including fairness controls confirm that the Luck effect remains robust while fairness ratings themselves are insignificant ([Table B7, Appendix B](#)). Moreover, *rich* participants send more under luck even when paired with other *rich* ([Table 9](#)),

<sup>8</sup> The corresponding results for trustworthiness, investment rate, and taking rate, estimated using subsample regressions by pairing type, are reported in [Appendix B](#) (see [Tables B4, B9, and B10](#)).

**Table 8**

Linear mixed effects regression results for trust game decisions (trust and trustworthiness), and Linear mixed effects probit regression results for investment in the stag-hunt game and taking in the prisoner's dilemma.

|                                                                                       | Trust               |                      | Trustworthiness      |                      | Investment Rate<br>Invest = 1;<br>Keep = 0 |                   | Taking Rate<br>Take = 1;<br>Not take = 0 |                   |
|---------------------------------------------------------------------------------------|---------------------|----------------------|----------------------|----------------------|--------------------------------------------|-------------------|------------------------------------------|-------------------|
|                                                                                       | (1)                 | (2)                  | (1)                  | (2)                  | (1)                                        | (2)               | (1)                                      | (2)               |
| <i>Fixed Effects</i>                                                                  |                     |                      |                      |                      |                                            |                   |                                          |                   |
| Intercept                                                                             | 38.988***<br>(1.59) | 41.802***<br>(13.35) | 39.428***<br>(1.145) | 24.978***<br>(9.671) | 0.504***<br>(0.081)                        | 0.87<br>(0.661)   | 0.392***<br>(0.074)                      | 0.299<br>(0.614)  |
| Luck treatment                                                                        | 5.037***<br>(1.88)  | 4.227**<br>(1.83)    | 0.44<br>(1.348)      | 0.273<br>(1.328)     | 0.062<br>(0.092)                           | 0.045<br>(0.092)  | 0.084<br>(0.085)                         | 0.081<br>(0.085)  |
| Rich status                                                                           | -11.24***<br>(1.87) | -13.493***<br>(1.96) | 3.249**<br>(1.347)   | 1.405<br>(1.420)     | 0.037<br>(0.092)                           | -0.074<br>(0.098) | -0.178**<br>(0.086)                      | -0.119<br>(0.092) |
| Encoding Score                                                                        |                     | 0.299*<br>(0.17)     |                      | 0.373***<br>(0.119)  |                                            | 0.009<br>(0.008)  |                                          | -0.012<br>(0.008) |
| Controls: demographics <sup>†</sup> , prior<br>experimental experience, SDO, &<br>RWA | No                  | Yes                  | No                   | Yes                  | No                                         | Yes               | No                                       | Yes               |
| Control for Order Effects                                                             | No                  | Yes                  | No                   | Yes                  | No                                         | Yes               | No                                       | Yes               |
| <i>Random Effects</i>                                                                 |                     |                      |                      |                      |                                            |                   |                                          |                   |
| Participant ID Variance                                                               | 511.69              | 466.15               | 257.58               | 242.16               | 0.40                                       | 0.33              | 0.29                                     | 0.24              |
| Participants / Obs. <sup>a</sup>                                                      | 692 / 1384          | 692 / 1384           | 692 / 1384           | 692 / 1384           | 692 / 1384                                 | 692 /<br>1384     | 692 / 1384                               | 692 /<br>1384     |
| AIC                                                                                   | 12,461.62           | 12,382.02            | 11,507.05            | 11,452.78            | 1726.60                                    | 1719.37           | 1825.16                                  | 1825.89           |
| Marginal R <sup>2</sup> / Conditional R <sup>2</sup>                                  | 0.049 /<br>0.746    | 0.111 /<br>0.746     | 0.008 /<br>0.748     | 0.060 /<br>0.748     | 0.001 /<br>0.285                           | 0.069 /<br>0.299  | 0.007 /<br>0.228                         | 0.161 /<br>0.326  |

*Notes:* For each outcome, specification (1) reports a baseline specification including the Luck treatment dummy and Rich status dummy. Specification (2) adds encoding score, demographic controls (age, gender, education, employment), prior experimental experience, SDO and RWA scores, and order controls.

Trust and trustworthiness are estimated using linear mixed effects regression models. Investment in the stag-hunt game (invest = 1; keep = 0) and taking in the prisoner's dilemma (take = 1; not take = 0) are estimated using linear mixed effects probit models.

"Luck" equals 1 for participants in the Luck treatments and 0 for those in the Merit treatments. "Rich status" equals 1 for participants endowed with 100 tokens and 0 for those endowed with 50. Figures in parentheses are standard errors. Each observation represents an independent decision from participants paired symmetrically in the equal and unequal treatments.

Significance levels:

\*\*\* p < 0.01.

\*\* p < 0.05.

\* p < 0.1.

**Table 9**

Results of Ordinary Least Square (OLS) regression models to explain trust in the trust game, categorised by interaction types.

|                                                                                                     | Poor-Poor<br>pairings |                      | Poor-Rich<br>pairings |                   | Rich-Rich<br>pairings |                     | Rich-Poor<br>pairings |                     |
|-----------------------------------------------------------------------------------------------------|-----------------------|----------------------|-----------------------|-------------------|-----------------------|---------------------|-----------------------|---------------------|
|                                                                                                     | (1)                   | (2)                  | (1)                   | (2)               | (1)                   | (2)                 | (1)                   | (2)                 |
| Intercept                                                                                           | 40.549***<br>(2.02)   | 74.624***<br>(18.67) | 37.692***<br>(2.047)  | 30.955<br>(19.04) | 28.038***<br>(2.111)  | -4.246<br>(30.186)  | 27.152***<br>(1.942)  | 5.596<br>(28.682)   |
| Luck treatment                                                                                      | 4.905*<br>(2.93)      | 3.729<br>(3.286)     | 4.611<br>(2.969)      | 4.337<br>(3.351)  | 5.278*<br>(2.867)     | 9.586***<br>(3.163) | 5.361**<br>(2.637)    | 8.241***<br>(3.005) |
| Encoding Score                                                                                      |                       | 0.014<br>(0.303)     |                       | -0.165<br>(0.309) |                       | 1.079***<br>(0.272) |                       | 0.642**<br>(0.258)  |
| Control for Demographics <sup>†</sup> , Prior<br>experience with experiments, SDO<br>and RWA scores | No                    | Yes                  | No                    | Yes               | No                    | Yes                 | No                    | Yes                 |
| Control for Order Effects                                                                           | No                    | Yes                  | No                    | Yes               | No                    | Yes                 | No                    | Yes                 |
| Obs.                                                                                                | 347                   | 347                  | 347                   | 347               | 345                   | 345                 | 345                   | 345                 |
| R <sup>2</sup> / R <sup>2</sup> adjusted                                                            | 0.008 /<br>0.005      | 0.082 /<br>0.017     | 0.007 /<br>0.004      | 0.070 /<br>0.003  | 0.010 /<br>0.007      | 0.172 /<br>0.113    | 0.012 /<br>0.009      | 0.119 /<br>0.056    |

*Notes:* Each column reports estimated coefficients from OLS regressions explaining first-mover trust, separately for each interaction type: *poor-poor*, *poor-rich*, *rich-rich*, and *rich-poor* pairings. "Luck treatment" equals 1 for participants in the Luck-based inequality condition and 0 for those in the Merit-based condition. Figures in parentheses are standard errors.

Significance levels:

\*\*\* p < 0.01.

\*\* p < 0.05.

\* p < 0.1.



where redistribution motives are irrelevant. We also do not find any significant differences between luck and merit in second-mover trustworthiness. If redistribution were a key motive, second movers (who have the greatest opportunity to equalise payoffs) should return a larger proportion of the tripled amount received.

We do observe limited evidence consistent with redistribution in one specific setting. In the merit treatment, among *rich* participants paired with *poor* partners, those who perceive the merit-based allocation as unfair tend to send more. In this setting, transfers decline as fairness perceptions increase from very unfair to very fair. However, this pattern appears only in this one case. In addition, the results in Table B8 in Appendix B do show that when the fairness perception variable is interacted with the Luck treatment, it renders the Luck treatment dummy insignificant. Overall, fairness perceptions may play a role in the transfers (as in the merit treatment), but we do not find conclusive evidence that redistributions motives are the predominant cause behind the higher trust in the luck treatment.

It is possible that higher first mover trust under luck than under merit may be explained by an entitlement effect. Prior research shows that people use their funds differently when it is earned compared to when it is unearned as they use different “mental accounts” for different income sources (Ambler and Godlonton, 2021; Cherry, 2001; Cherry et al., 2002; Jakiela, 2015; Umer et al., 2022). People feel more entitled to their earned income potentially leading to less prosocial behaviour compared to unearned income (Bulte et al., 2016; Cherry, 2001; Cherry et al., 2002; Jakiela, 2015). While participants perceive a merit-based system to be fairer, this perception does not translate to higher trust, possibly because participants under merit may feel justified in keeping their earned tokens rather than transferring since that makes the trustor vulnerable to betrayal by the recipient, who can appropriate any surplus created (Bohnet et al., 2008; Bohnet and Zeckhauser, 2004).

This result that trust is higher under luck was robust to potential cultural variation among participants drawn from 34 countries. The sample included substantial representation from Mexico, Poland, Portugal, and South Africa, with fewer participants from Western Europe or the United States. As country of birth captured participants’ backgrounds, we examined whether our key result (i.e., higher trust under luck) was driven by specific national or regional affiliation. We ran a series of “leave-one-out” regressions, excluding one country or subregional groups (Southern Africa, Western/Eastern Europe, and Latin America) at a time. The results remained stable across all specifications, indicating that no single country or region drove the observed treatment effect.

#### 4.3. Fairness perceptions

In the final questionnaire, participants assessed the fairness of their treatment group’s endowment allocation method, both before and after they learned about the other methods. Looking at the top panel of Table 10, we can see that participants in the *Unequal-Merit* treatments rated their method as fairest, followed by the *Equal* and the *Unequal-Luck* treatments. Non-parametric Wilcoxon rank sum tests reveal that learning about other methods reduced the perceived fairness of luck (from a rating of 3.54 to 3.36), while the ratings for merit (3.67 to 3.76) and equal (3.61 to 3.73) methods improved. This highlights that having a reference point amplifies the perceived unfairness of luck.

**Table 10**  
Average fairness perception score of the three endowment allocation methods.

| Treatment Group                                             | Obs. | Average fairness score (Before knowledge about other methods) | Average fairness score (After knowledge about other methods) | Wilcoxon Rank Sum Test p-value |
|-------------------------------------------------------------|------|---------------------------------------------------------------|--------------------------------------------------------------|--------------------------------|
| Equal                                                       | 352  | 3.605<br>(0.979)                                              | 3.727<br>(1.024)                                             | 0.036                          |
| Luck                                                        | 352  | 3.540<br>(1.053)                                              | 3.358<br>(1.039)                                             | < 0.001                        |
| Merit                                                       | 340  | 3.674<br>(1.000)                                              | 3.762<br>(1.036)                                             | 0.041                          |
| Wilcoxon Rank Sum Test p-value comparing between treatments |      |                                                               |                                                              |                                |
|                                                             |      | Before knowledge about other methods                          | After knowledge about other methods                          |                                |
| Equal vs. Luck                                              |      | 0.345                                                         | < 0.001                                                      |                                |
| Equal vs. Merit                                             |      | 0.275                                                         | 0.599                                                        |                                |
| Luck vs. Merit                                              |      | 0.054                                                         | < 0.001                                                      |                                |

*Notes:* The fairness perception rating ranges from 0 (very unfair) to 5 (very fair). The Wilcoxon Rank-Sum test p-values indicate whether differences in perceived fairness are statistically significant within each treatment (before vs after) and between treatments (Equal vs Luck, Equal vs Merit, Luck vs Merit). Figures in parentheses are standard deviations.

Prior to learning about different allocation methods, there are still significant differences in fairness ratings between merit and luck ( $p$ -value = 0.054) but not for the other comparisons. After learning about other allocations methods, both merit and equal are rated higher than luck with highly significant differences in both cases ( $p$ -value < 0.001). But differences were not significant between merit and equal, implying that there is no discernible fairness perception gap between an effort-based strategy and an egalitarian one among participants in a lab setting.

We further measured participants' fairness preferences for different allocation methods by asking them to choose the fairest method between equal, luck-based, merit-based, and uncertain options in the questionnaire.<sup>9</sup> Overall, participants have distinct fairness preferences for different endowment allocation processes ( $\chi^2 = 430.12$ ,  $df = 3$ ,  $p$ -value < 0.001). Approximately 46 per cent of the participants are meritocratic in the sense that they consider merit to be the fairest allocation method. About 35 per cent are egalitarians, preferring equality while 11 per cent and 8 per cent respectively prefer luck (luck acceptors) or uncertainty (uncertainty-acceptors). Table B11 in Appendix B shows the percentage of participants who selected each option grouped by treatment.

Incorporating this heterogeneity in fairness preferences adds considerable nuance to our results. The evidence suggests that differences in fairness preferences lead to interpretable differences in behaviour. For instance, earlier we reported that trustworthiness is lower in unequal societies for participants in general. When we disaggregated the results by fairness preference type, we found that egalitarians (participants who viewed equality as the fairest allocation method) were notably more sensitive to inequality than meritocrats (who viewed merit as fairest) and accordingly reciprocated less under unequal conditions. A full analysis of these interactions, including how fairness preferences interact with endowment status and treatment type, lies beyond the scope of the current paper. We plan to report these results in follow-up work.

## 5. Conclusion

In this study, we use economic experiments to manipulate the level and the source of inequality among participants and measure their subsequent behaviour using three economic games, trust game, stag-hunt game and the prisoner's dilemma game, that are good proxies for measuring social capital. This allows us to test the causal effect of inequality, as well as to compare the differential impact of luck and merit-based inequality on social capital. Our findings indicate that inequality has a negative effect on some, but not all, dimensions of social capital. Inequality reduces trustworthiness and the propensity to coordinate to the payoff dominant outcome in the stag-hunt game. Overall, these findings are broadly consistent with Yang and Konrath's (2023) meta-analysis that identified a small but significant negative link between inequality and prosocial behaviour, a proxy for social capital (Brooks, 2005; Wang and Graddy, 2008).

On comparing the two sources of inequality, we find that, by and large, they have little impact on social capital. One possible explanation is a gap between stated and revealed preferences, a well-documented pattern in economics (List and Gallet, 2001). Participants may view merit-based inequality as fairer in surveys, but when real money is on the line, their behaviour does not reflect these stated beliefs. This disconnect may account for the lack of behavioural differences across games. However, we do find evidence of increased trust under luck compared to merit. This is contrary to our hypothesis and is especially interesting in the light of our other finding that participants perceive merit-based allocation to be fairer than luck. We have argued that this finding is likely due to an entitlement effect rather than redistributive motives.

Overall, this study highlights the interplay between the source of inequality, perceived fairness, and its impact on social capital. While merit-based systems are perceived as fairer, they may undermine trust, challenging assumptions about fairness and social capital. Future research should investigate why individuals trust more in luck-based systems despite perceiving them as less fair. This could involve examining psychological factors like entitlement and perceived fairness using validated scales like the Support for Economic Inequality Scale (SEIS). Comparative studies that incorporate other economic games, like public goods and dictator games, can verify our findings.

## Authors' contributions

All authors were equally involved in the design of experiments. EL programmed the experiment in o-Tree and, along with AR conducted data collection via Prolific. SS handled the data analysis with input from EL, AR and AC. SS and AC were responsible for writing the paper. All authors reviewed and approved the final manuscript.

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## Ethics approval and consent

The experiment was approved by the University of Auckland Human Participants Ethics Committee (Ref. No.: UAHPEC23602).

<sup>9</sup> Recall that the uncertain allocation method simply means a 50 per cent chance of the endowment being determined based on luck and a 50 per cent chance of it being determined based on merit.

## Data availability

We have uploaded the data to [Mendeley Data](#).

## Declaration of competing interest

The authors declare that they have no conflicts of interest with respect to their authorship or the publication of this article.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jebo.2026.107588](https://doi.org/10.1016/j.jebo.2026.107588).

## Appendix A. Summary statistics and balance Tables

| ‘Investment Rate’ in the Stag-hunt Game |      |                                             |                                             |                                             |                                             |
|-----------------------------------------|------|---------------------------------------------|---------------------------------------------|---------------------------------------------|---------------------------------------------|
| Treatment Group                         | Obs. | Poor-Poor<br>% of Participants who ‘Invest’ | Rich-Rich<br>% of Participants who ‘Invest’ | Poor-Rich<br>% of Participants who ‘Invest’ | Rich-Poor<br>% of Participants who ‘Invest’ |
| <b>Equal</b>                            | 352  | 83.44 %<br>(0.37)                           | 81.48 %<br>(0.39)                           | –                                           | –                                           |
| <b>Unequal-Luck</b>                     | 352  | 74.55 %<br>(0.44)                           | 74.87 %<br>(0.43)                           | 61.21 %<br>(0.49)                           | 62.03 %<br>(0.49)                           |
| <b>Unequal-Merit</b>                    | 340  | 69.23 %<br>(0.46)                           | 73.42 %<br>(0.44)                           | 62.09 %<br>(0.49)                           | 60.76 %<br>(0.49)                           |

| ‘Taking Rate’ in the Prisoner’s Dilemma Game |      |                                           |                                           |                                           |                                           |
|----------------------------------------------|------|-------------------------------------------|-------------------------------------------|-------------------------------------------|-------------------------------------------|
| Treatment Group                              | Obs. | Poor-Poor<br>% of Participants who ‘Take’ | Rich-Rich<br>% of Participants who ‘Take’ | Poor-Rich<br>% of Participants who ‘Take’ | Rich-Poor<br>% of Participants who ‘Take’ |
| <b>Equal</b>                                 | 352  | 53.99 %<br>(0.50)                         | 58.73 %<br>(0.49)                         | –                                         | –                                         |
| <b>Unequal-Luck</b>                          | 352  | 58.79 %<br>(0.49)                         | 58.29 %<br>(0.49)                         | 77.58 %<br>(0.42)                         | 57.22 %<br>(0.50)                         |
| <b>Unequal-Merit</b>                         | 340  | 48.90 %<br>(0.50)                         | 56.33 %<br>(0.50)                         | 73.08 %<br>(0.44)                         | 63.29 %<br>(0.48)                         |

| Trust in the Trust Game |      |                                  |                                  |                                  |                                  |
|-------------------------|------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| Treatment Group         | Obs. | Poor-Poor<br>Send Rate<br>(in %) | Rich-Rich<br>Send Rate<br>(in %) | Poor-Rich<br>Send Rate<br>(in %) | Rich-Poor<br>Send Rate<br>(in %) |
| <b>Equal</b>            | 352  | 43.37 %<br>(28.40)               | 31.96 %<br>(25.35)               | –                                | –                                |
| <b>Unequal-Luck</b>     | 352  | 45.45 %<br>(28.85)               | 33.32 %<br>(28.32)               | 42.3 %<br>(29.40)                | 32.51 %<br>(25.77)               |
| <b>Unequal-Merit</b>    | 340  | 40.55 %<br>(25.72)               | 28.04 %<br>(24.25)               | 37.69 %<br>(25.90)               | 27.15 %<br>(22.69)               |

| Trustworthiness in the Trust Game |      |                                    |                                    |                                    |                                    |
|-----------------------------------|------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| Treatment Group                   | Obs. | Poor-Poor<br>Return Rate<br>(in %) | Rich-Rich<br>Return Rate<br>(in %) | Poor-Rich<br>Return Rate<br>(in %) | Rich-Poor<br>Return Rate<br>(in %) |

(continued on next page)

(continued)

| Trustworthiness in the Trust Game |      |                              |                              |                              |                              |
|-----------------------------------|------|------------------------------|------------------------------|------------------------------|------------------------------|
| Treatment Group                   | Obs. | Poor-Poor Return Rate (in %) | Rich-Rich Return Rate (in %) | Poor-Rich Return Rate (in %) | Rich-Poor Return Rate (in %) |
| <b>Equal</b>                      | 352  | 43.88 %<br>(18.18)           | 39.55 %<br>(17.29)           | –                            | –                            |
| <b>Unequal-Luck</b>               | 352  | 41.74 %<br>(17.52)           | 41.52 %<br>(20.15)           | 38.94 %<br>(17.24)           | 43.88 %<br>(22.17)           |
| <b>Unequal-Merit</b>              | 340  | 40.18 %<br>(16.59)           | 42.94 %<br>(18.62)           | 37.82 %<br>(17.55)           | 43.41 %<br>(20.74)           |

The tables below present balance tests on key demographics. The results indicate no significant differences in the distribution of respondents across the three methods of endowment allocation: luck, merit, and equal.

| Balance tests on key demographics |            |              |               |         |
|-----------------------------------|------------|--------------|---------------|---------|
| Covariates                        | Equal      | Unequal-Luck | Unequal-Merit | p-value |
| N                                 | 352        | 352          | 340           |         |
| <b>Age</b>                        | 27.06      | 26.92        | 26.80         | 0.919   |
| [mean (SD)]                       | (8.89)     | (8.54)       | (8.10)        |         |
| <b>Female</b>                     | 183        | 189          | 206           | 0.056   |
| [N (%)]                           | (52.0 %)   | (53.7 %)     | (60.6 %)      |         |
| <b>Education</b>                  | 3.52       | 3.62         | 3.59          | 0.545   |
| [mean (SD)]                       | (1.19)     | (1.17)       | (1.11)        |         |
| <b>Employment</b>                 | 2.38       | 2.37         | 2.48          | 0.531   |
| [mean (SD)]                       | (1.44)     | (1.46)       | (1.47)        |         |
| <b>Country of Birth</b>           |            |              |               | 0.792   |
| [N (%)]                           |            |              |               |         |
| Argentina                         | 0<br>(0.0) | 2<br>(0.6)   | 1 (0.3)       |         |
| Australia                         | 2<br>(0.6) | 1<br>(0.3)   | 0 (0.0)       |         |
| Austria                           | 0 (0.0)    | 1<br>(0.3)   | 0 (0.0)       |         |
| Belgium                           | 3 (0.9)    | 3 (0.9)      | 2 (0.6)       |         |
| Bolivia                           | 0 (0.0)    | 1 (0.3)      | 0 (0.0)       |         |
| Brazil                            | 2 (0.6)    | 2 (0.6)      | 1 (0.3)       |         |
| Canada                            | 4 (1.1)    | 4 (1.1)      | 5 (1.5)       |         |
| Chile                             | 6 (1.7)    | 6 (1.7)      | 5 (1.5)       |         |
| China                             | 0 (0.0)    | 2 (0.6)      | 0 (0.0)       |         |
| Colombia                          | 0 (0.0)    | 0 (0.0)      | 1 (0.3)       |         |
| Czech Republic                    | 4 (1.1)    | 3 (0.9)      | 3 (0.9)       |         |
| Denmark                           | 1 (0.3)    | 0 (0.0)      | 0 (0.0)       |         |
| Estonia                           | 2 (0.6)    | 1 (0.3)      | 4 (1.2)       |         |
| Finland                           | 4 (1.1)    | 4 (1.1)      | 3 (0.9)       |         |
| France                            | 4 (1.1)    | 4 (1.1)      | 5 (1.5)       |         |
| Germany                           | 3 (0.9)    | 4 (1.1)      | 2 (0.6)       |         |
| Greece                            | 7 (2.0)    | 11 (3.1)     | 13 (3.8)      |         |
| Hungary                           | 4 (1.1)    | 11 (3.1)     | 9 (2.6)       |         |
| Iceland                           | 0 (0.0)    | 0 (0.0)      | 1 (0.3)       |         |
| India                             | 1 (0.3)    | 0 (0.0)      | 1 (0.3)       |         |
| Iran                              | 1 (0.3)    | 0 (0.0)      | 0 (0.0)       |         |
| Israel                            | 2 (0.6)    | 2 (0.6)      | 1 (0.3)       |         |
| Italy                             | 27 (7.7)   | 17 (4.8)     | 26 (7.6)      |         |
| Latvia                            | 2 (0.6)    | 4 (1.1)      | 2 (0.6)       |         |
| Mexico                            | 37 (10.5)  | 34 (9.7)     | 29 (8.5)      |         |
| Namibia                           | 0 (0.0)    | 1 (0.3)      | 0 (0.0)       |         |
| Netherlands                       | 5 (1.4)    | 6 (1.7)      | 1 (0.3)       |         |
| New Zealand                       | 1 (0.3)    | 0 (0.0)      | 0 (0.0)       |         |
| Nigeria                           | 0 (0.0)    | 0 (0.0)      | 1 (0.3)       |         |
| Norway                            | 0 (0.0)    | 2 (0.6)      | 0 (0.0)       |         |
| Pakistan                          | 0 (0.0)    | 0 (0.0)      | 1 (0.3)       |         |
| Poland                            | 49 (13.9)  | 45 (12.8)    | 47 (13.8)     |         |
| Portugal                          | 61 (17.3)  | 59 (16.8)    | 62 (18.2)     |         |
| Qatar                             | 1 (0.3)    | 0 (0.0)      | 0 (0.0)       |         |
| Republic of Ireland               | 2 (0.6)    | 0 (0.0)      | 1 (0.3)       |         |
| Romania                           | 2 (0.6)    | 0 (0.0)      | 1 (0.3)       |         |

(continued on next page)

(continued)

| Balance tests on key demographics |           |              |               |         |
|-----------------------------------|-----------|--------------|---------------|---------|
| Covariates                        | Equal     | Unequal-Luck | Unequal-Merit | p-value |
| N                                 | 352       | 352          | 340           |         |
| Slovakia                          | 0 (0.0)   | 0 (0.0)      | 1 (0.3)       |         |
| Slovenia                          | 4 (1.1)   | 1 (0.3)      | 1 (0.3)       |         |
| Somalia                           | 0 (0.0)   | 0 (0.0)      | 1 (0.3)       |         |
| South Africa                      | 77 (21.9) | 85 (24.1)    | 77 (22.6)     |         |
| South Sudan                       | 0 (0.0)   | 0 (0.0)      | 1 (0.3)       |         |
| Spain                             | 13 (3.7)  | 13 (3.7)     | 9 (2.6)       |         |
| Sweden                            | 3 (0.9)   | 1 (0.3)      | 1 (0.3)       |         |
| Switzerland                       | 3 (0.9)   | 0 (0.0)      | 1 (0.3)       |         |
| Syria                             | 1 (0.3)   | 0 (0.0)      | 0 (0.0)       |         |
| Taiwan                            | 1 (0.3)   | 0 (0.0)      | 0 (0.0)       |         |
| Turkey                            | 0 (0.0)   | 1 (0.3)      | 0 (0.0)       |         |
| Ukraine                           | 1 (0.3)   | 1 (0.3)      | 0 (0.0)       |         |
| United Arab Emirates              | 0 (0.0)   | 1 (0.3)      | 0 (0.0)       |         |
| United Kingdom                    | 7 (2.0)   | 16 (4.5)     | 16 (4.7)      |         |
| United States                     | 1 (0.3)   | 3 (0.9)      | 2 (0.6)       |         |
| Zimbabwe                          | 4 (1.1)   | 0 (0.0)      | 2 (0.6)       |         |

## Appendix B. Other Tables and Figures

Tables B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, Fig. B1.

**Table B1**

Payoff matrices of the stag-hunt game for all combinations of pairings between participants based on their endowment.

|               |                               | Participant 2                |          |         |                               |          |          |
|---------------|-------------------------------|------------------------------|----------|---------|-------------------------------|----------|----------|
|               |                               | Poor [Endowment = 50 tokens] |          |         | Rich [Endowment = 100 tokens] |          |          |
| Participant 1 | Poor [Endowment = 50 tokens]  | Invest                       | 110, 110 | 20, 80  | Invest                        | 140, 190 | 20, 130  |
|               |                               | Keep                         | 80, 20   | 50, 50  | Keep                          | 110, 40  | 50, 100  |
|               | Rich [Endowment = 100 tokens] | Invest                       | 190, 140 | 40, 110 | Invest                        | 220, 220 | 40, 160  |
|               |                               | Keep                         | 130, 20  | 100, 50 | Keep                          | 160, 40  | 100, 100 |

**Table B2**

Payoff matrices of the prisoner's dilemma for all combinations of pairings between participants based on their endowment.

|               |                               | Participant 2                |        |         |                               |         |          |
|---------------|-------------------------------|------------------------------|--------|---------|-------------------------------|---------|----------|
|               |                               | Poor [Endowment = 50 tokens] |        |         | Rich [Endowment = 100 tokens] |         |          |
| Participant 1 | Poor [Endowment = 50 tokens]  | Take                         | 35, 35 | 65, 20  | Take                          | 50, 55  | 80, 40   |
|               |                               | Not take                     | 20, 65 | 50, 50  | Not take                      | 20, 115 | 50, 100  |
|               | Rich [Endowment = 100 tokens] | Take                         | 55, 50 | 115, 20 | Take                          | 70, 70  | 130, 40  |
|               |                               | Not take                     | 40, 80 | 100, 50 | Not take                      | 40, 130 | 100, 100 |

**Table B3**

Participant dropouts at different experimental stages by treatment group.

| Stage                                   | Equal Treatments | Unequal-Merit Treatment | Unequal-Luck Treatment | Total Dropouts |
|-----------------------------------------|------------------|-------------------------|------------------------|----------------|
| Dropouts during Initial Instructions    | 14               | 14                      | 16                     | 44             |
| Dropouts during Stag-Hunt Game          | 0                | 5                       | 4                      | 9              |
| Dropouts during Prisoner's Dilemma Game | 2                | 2                       | 3                      | 7              |
| Dropouts during Trust Game*             | 29               | 33                      | 43                     | 105            |
| Dropouts during Final Questionnaire     | 0                | 1                       | 2                      | 3              |
| Total Dropouts                          | 45               | 55                      | 68                     | 168            |

\* The high dropout rate during the trust game was primarily driven by participants' failure to pass the validation check question. This game also had higher cognitive demands and a longer completion time than the stag-hunt game and the prisoner's dilemma, which likely contributed to additional attrition. Importantly, all games were presented in random order, ruling out systematic ordering effects. Moreover, since participants learned their final earnings only at the end of the experiment, selective dropout based on performance is unlikely.

**Table B4**

Results of Ordinary Least Square (OLS) regression models to explain trustworthiness in the trust game, categorised by pairing types.

|                                                                                 | Poor-Poor pairings   |                      | Poor-Rich pairings   |                      | Rich-Rich pairings   |                     | Rich-Poor pairings   |                    |
|---------------------------------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|---------------------|----------------------|--------------------|
|                                                                                 | (1)                  | (2)                  | (1)                  | (2)                  | (1)                  | (2)                 | (1)                  | (2)                |
| Intercept                                                                       | 40.176***<br>(1.263) | 24.431**<br>(11.624) | 37.817***<br>(1.290) | 25.454**<br>(12.024) | 42.940***<br>(1.548) | 41.316*<br>(22.540) | 43.409***<br>(1.713) | 38.111<br>(25.103) |
| Luck dummy                                                                      | 1.568<br>(1.832)     | 0.075<br>(2.046)     | 1.127<br>(1.871)     | −0.387<br>(2.116)    | −1.422<br>(2.103)    | 1.266<br>(2.362)    | 0.468<br>(2.326)     | 3.308<br>(2.630)   |
| Encoding Score                                                                  |                      | 0.275<br>(0.189)     |                      | 0.315<br>(0.195)     |                      | 0.484**<br>(0.203)  |                      | 0.577**<br>(0.226) |
| Control for Demographics, Prior experience with experiments, SDO and RWA scores | No                   | Yes                  | No                   | Yes                  | No                   | Yes                 | No                   | Yes                |
| Control for Order Effects                                                       | No                   | Yes                  | No                   | Yes                  | No                   | Yes                 | No                   | Yes                |
| Obs.                                                                            | 347                  | 347                  | 347                  | 347                  | 345                  | 345                 | 345                  | 345                |
| R <sup>2</sup> / R <sup>2</sup> adjusted                                        | 0.002 /<br>0.001     | 0.085 /<br>0.020     | 0.001<br>/0.002      | 0.060 /<br>0.007     | 0.001 /<br>0.002     | 0.135 /<br>0.073    | 0.000 /<br>0.003     | 0.122 /<br>0.059   |

\*\*\* p &lt; 0.01.

\*\* p &lt; 0.05.

\* p &lt; 0.1.

^ Demographic variables include age, gender, education and employment Note: Figures in parentheses are standard errors.

**Table B5**

Multiple Hypothesis Correction for unequal treatment effect across four outcomes.

|                | Trust |       | Trustworthiness |       | Investment Rate<br>Invest = 1;<br>Keep = 0 |       | Taking Rate<br>Take = 1;<br>Not take = 0 |       |
|----------------|-------|-------|-----------------|-------|--------------------------------------------|-------|------------------------------------------|-------|
|                | (1)   | (2)   | (1)             | (2)   | (1)                                        | (2)   | (1)                                      | (2)   |
| p-value (Raw)  | 0.654 | 0.936 | 0.044           | 0.089 | 0.001                                      | 0.001 | 0.785                                    | 0.477 |
| p-value (Bonf) | 1.0   | 1.0   | 0.175           | 0.356 | 0.003                                      | 0.003 | 1.000                                    | 1.0   |
| p-value (BH)   | 0.785 | 0.936 | 0.087           | 0.178 | 0.003                                      | 0.003 | 0.785                                    | 0.636 |

Notes: The table reports raw and adjusted p-values for the Unequal treatment effect across the four main outcomes in each regression specification. Columns (1) and (2) correspond to the two specifications in Table 5. Bonf denotes Bonferroni-adjusted p-values controlling the family-wise error rate, and BH denotes Benjamini-Hochberg-adjusted p-values controlling the false discovery rate. Corrected p-values are truncated at 1.0 where applicable.

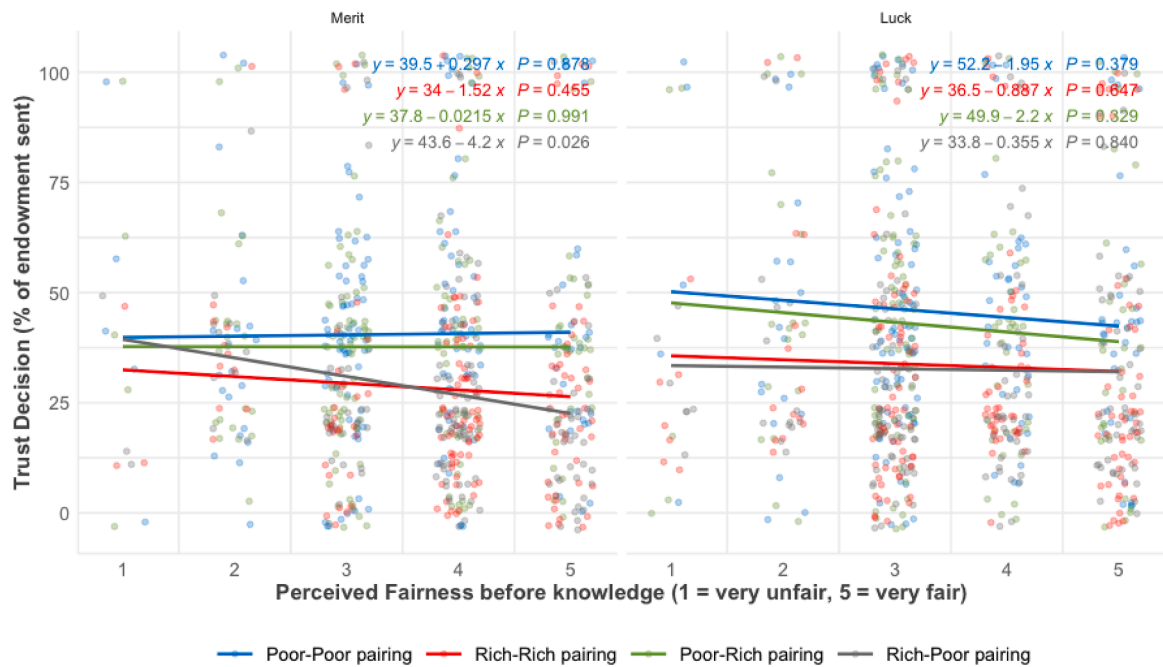
**Table B6**

Multiple Hypothesis Correction for luck treatment effect across four outcomes.

|                | Trust |       | Trustworthiness |       | Investment Rate<br>Invest = 1;<br>Keep = 0 |       | Taking Rate<br>Take = 1;<br>Not take = 0 |       |
|----------------|-------|-------|-----------------|-------|--------------------------------------------|-------|------------------------------------------|-------|
|                | (1)   | (2)   | (1)             | (2)   | (1)                                        | (2)   | (1)                                      | (2)   |
| p-value (Raw)  | 0.007 | 0.021 | 0.744           | 0.837 | 0.504                                      | 0.625 | 0.327                                    | 0.340 |
| p-value (Bonf) | 0.028 | 0.086 | 1.0             | 1.0   | 1.0                                        | 1.0   | 1.0                                      | 1.0   |
| p-value (BH)   | 0.028 | 0.086 | 0.744           | 0.837 | 0.672                                      | 0.833 | 0.654                                    | 0.680 |

Notes: The table reports raw and adjusted p-values for the Luck treatment effect across the four main outcomes in each regression specification. Columns (1) and (2) correspond to the two specifications in Table 8. Bonf denotes Bonferroni-adjusted p-values controlling the family-wise error rate, and BH denotes Benjamini-Hochberg-adjusted p-values controlling the false discovery rate. Corrected p-values are truncated at 1.0 where applicable.





**Fig. B1.** Scatter plot showing the relationship between perceived fairness before knowledge of other endowment allocation methods and initial transfer rate in the trust game.

**Table B7**

Mixed-effects regression results for trust decisions in the trust game with fairness assessments included as additional controls.

| Regressors                                                                       | (1)                   | (2)                    | (3)                    |
|----------------------------------------------------------------------------------|-----------------------|------------------------|------------------------|
| Intercept                                                                        | 41.293 ***<br>(3.684) | 43.344 ***<br>(3.618)  | 45.294 ***<br>(13.479) |
| Luck treatment                                                                   | 4.015 **<br>(1.916)   | 4.846 ***<br>(1.878)   | 4.001 **<br>(1.834)    |
| Rich status                                                                      |                       | -10.848 ***<br>(1.895) | -13.100 ***<br>(1.969) |
| Fairness perception score                                                        | -2.049 **<br>(0.932)  | -1.235<br>(0.921)      | -1.519 *<br>(0.902)    |
| Encoding Score                                                                   |                       |                        | 0.323 *<br>(0.165)     |
| Control for Demographics , Prior experience with experiments, SDO and RWA scores | No                    | No                     | Yes                    |
| Random Effects                                                                   |                       |                        |                        |
| Residual variance $\sigma^2$                                                     | 186.92                | 186.92                 | 186.92                 |
| Random intercept variance $\tau_{00}$                                            | 538.71 participant_ID | 510.12 participant_ID  | 463.87 participant_ID  |
| Intraclass Correlation Coefficient (ICC)                                         | 0.74                  | 0.73                   | 0.71                   |
| Participants                                                                     | 692 participant_ID    | 692 participant_ID     | 692 participant_ID     |
| Observations                                                                     | 1384                  | 1384                   | 1384                   |
| Marginal $R^2$ / Conditional $R^2$                                               | 0.012 / 0.746         | 0.051 / 0.746          | 0.114 / 0.746          |
| AIC                                                                              | 12,493.159            | 12,460.151             | 12,379.645             |

\*\*\*  $p < 0.01$ .

\*\*  $p < 0.05$ .

\*  $p < 0.1$ .

^ Demographic variables include age, gender, education and employment.

**Table B8**

Mixed-effects regression results for trust decisions in the trust game with fairness assessments included as controls interacted with the luck treatment dummy.

| Regressors           | (1)<br>Estimates      | (2)<br>Estimates      | (3)<br>Estimates       |
|----------------------|-----------------------|-----------------------|------------------------|
| Intercept            | 42.626 ***<br>(5.200) | 43.246 ***<br>(5.082) | 46.494 ***<br>(13.806) |
| Luck treatment dummy | 1.566<br>(7.012)      | 5.028<br>(6.879)      | 1.418<br>(6.706)       |

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Table B8 (continued)

| Regressors                                                                       | (1)<br>Estimates      | (2)<br>Estimates       | (3)<br>Estimates       |
|----------------------------------------------------------------------------------|-----------------------|------------------------|------------------------|
| Fairness perception score                                                        | −2.412 *<br>(1.366)   | −1.208<br>(1.351)      | −1.913<br>(1.335)      |
| Fairness perception score x Luck dummy                                           | 0.678<br>(1.868)      | −0.050<br>(1.830)      | 0.714<br>(1.784)       |
| Rich status                                                                      |                       | −10.852 ***<br>(1.899) | −13.064 ***<br>(1.970) |
| Encoding Score                                                                   |                       |                        | 0.328 **<br>(0.165)    |
| Control for Demographics , Prior experience with experiments, SDO and RWA scores | No                    | No                     | Yes                    |
| Random Effects                                                                   |                       |                        |                        |
| Residual variance $\sigma^2$                                                     | 186.92                | 186.92                 | 186.92                 |
| Random intercept variance $\tau_{00}$                                            | 538.59 participant_ID | 510.12 participant_ID  | 463.74 participant_ID  |
| Intraclass Correlation Coefficient (ICC)                                         | 0.74                  | 0.73                   | 0.71                   |
| Participants                                                                     | 692 participant_ID    | 692 participant_ID     | 692 participant_ID     |
| Observations                                                                     | 1384                  | 1384                   | 1384                   |
| Marginal R <sup>2</sup> / Conditional R <sup>2</sup>                             | 0.012 / 0.746         | 0.051 / 0.746          | 0.114 / 0.746          |
| AIC                                                                              | 12,491.938            | 12,459.101             | 12,378.476             |

\*\*\* p &lt; 0.01.

\*\* p &lt; 0.05.

\* p &lt; 0.1.

^ Demographic variables include age, gender, education and employment.

Table B9

Results of Probit regression models to explain ‘investment rate’ in the stag-hunt game, categorised by pairing types.

|                                                                                  | Poor-Poor pairings |                     | Poor-Rich pairings |                     | Rich-Rich pairings |                   | Rich-Poor pairings |                   |
|----------------------------------------------------------------------------------|--------------------|---------------------|--------------------|---------------------|--------------------|-------------------|--------------------|-------------------|
|                                                                                  | (1)                | (2)                 | (1)                | (2)                 | (1)                | (2)               | (1)                | (2)               |
| Intercept                                                                        | 0.502<br>***       | 0.393<br>(0.097)    | 0.308<br>***       | 1.820 **<br>(0.095) | 0.625<br>***       | 4.964<br>(0.107)  | 0.273<br>***       | 4.854<br>(0.101)  |
| Luck dummy                                                                       |                    | 0.031<br>(0.144)    | −0.023<br>(0.137)  | −0.112<br>(0.159)   | 0.045<br>(0.146)   | 0.021<br>(0.176)  | 0.033<br>(0.138)   | −0.013<br>(0.164) |
| Encoding Score                                                                   |                    | 0.032 **<br>(0.016) |                    | 0.029 **<br>(0.015) |                    | −0.006<br>(0.015) |                    | −0.005<br>(0.014) |
| Control for Demographics , Prior experience with experiments, SDO and RWA scores | No                 | Yes                 | No                 | Yes                 | No                 | Yes               | No                 | Yes               |
| Control for Order Effects                                                        | No                 | Yes                 | No                 | Yes                 | No                 | Yes               | No                 | Yes               |
| Obs.                                                                             | 347                | 347                 | 347                | 347                 | 345                | 345               | 345                | 345               |
| Nagelkerke R <sup>2</sup>                                                        | 0.005              | 0.16                | 0                  | 0.145               | 0                  | 0.124             | 0                  | 0.101             |
| AIC                                                                              | 415.876            | 419.995             | 465.933            | 470.759             | 397.844            | 411.521           | 463.962            | 481.369           |

\*\*\* p &lt; 0.01.

\*\* p &lt; 0.05

\* p &lt; 0.1.

^ Demographic variables include age, gender, education and employment

Note: Figures in parentheses are standard errors.

Table B10

Results of Probit regression models to explain ‘taking rate’ in the prisoner’s dilemma, categorised by pairing types.

|                                                                                  | Poor-Poor pairings |                        | Poor-Rich pairings |                  | Rich-Rich pairings |                   | Rich-Poor pairings |                   |
|----------------------------------------------------------------------------------|--------------------|------------------------|--------------------|------------------|--------------------|-------------------|--------------------|-------------------|
|                                                                                  | (1)                | (2)                    | (1)                | (2)              | (1)                | (2)               | (1)                | (2)               |
| Intercept                                                                        | −0.028             | −0.402<br>(0.093)      | 0.615<br>***       | 1.488<br>(0.100) | 0.159              | 5.572<br>(0.100)  | 0.340<br>***       | 4.259<br>(0.102)  |
| Luck dummy                                                                       |                    | 0.361 **<br>(0.135)    | 0.143<br>(0.147)   | 0.112<br>(0.176) | 0.05<br>(0.136)    | −0.023<br>(0.161) | −0.158<br>(0.137)  | −0.191<br>(0.165) |
| Encoding Score                                                                   |                    | −0.028<br>*<br>(0.015) |                    | 0.004<br>(0.017) |                    | −0.021<br>(0.014) |                    | −0.011<br>(0.014) |
| Control for Demographics , Prior experience with experiments, SDO and RWA scores | No                 | Yes                    | No                 | Yes              | No                 | Yes               | No                 | Yes               |
| Control for Order Effects                                                        | No                 | Yes                    | No                 | Yes              | No                 | Yes               | No                 | Yes               |
| Obs.                                                                             | 347                | 347                    | 347                | 347              | 345                | 345               | 345                | 345               |
| Nagelkerke R <sup>2</sup>                                                        | 0.013              | 0.126                  | 0.004              | 0.153            | 0.001              | 0.095             | 0.005              | 0.156             |
| AIC                                                                              | 479.833            | 492.821                | 391.662            | 398.793          | 474.57             | 493.335           | 467.06             | 470.073           |

\*\*\*  $p < 0.01$ .\*\*  $p < 0.05$ .\*  $p < 0.1$ .

^ Demographic variables include age, gender, education and employment

Note: Figures in parentheses are standard errors.

**Table B11**

Participants' response to which endowment allocation process do they find to be the "fairest" by treatment and endowment status.

| Treatment        | Endowment | Obs. | Which endowment allocation process is the fairest? |         |         |           |
|------------------|-----------|------|----------------------------------------------------|---------|---------|-----------|
|                  |           |      | Egalitarian                                        | Luck    | Merit   | Uncertain |
| Equal            | Poor      | 163  | 38.65 %                                            | 7.98 %  | 47.85 % | 5.52 %    |
|                  | Rich      | 189  | 44.44 %                                            | 4.76 %  | 46.03 % | 4.76 %    |
|                  | All       | 352  | 41.76 %                                            | 6.25 %  | 46.88 % | 5.11 %    |
| Unequal-Luck     | Poor      | 165  | 39.39 %                                            | 16.36 % | 35.15 % | 9.09 %    |
|                  | Rich      | 187  | 35.83 %                                            | 19.79 % | 35.29 % | 9.09 %    |
|                  | All       | 352  | 37.50 %                                            | 18.18 % | 35.23 % | 9.09 %    |
| Unequal-Merit    | Poor      | 182  | 31.32 %                                            | 12.09 % | 45.60 % | 10.99 %   |
|                  | Rich      | 158  | 16.46 %                                            | 5.70 %  | 70.25 % | 7.59 %    |
|                  | All       | 340  | 24.41 %                                            | 9.12 %  | 57.06 % | 9.41 %    |
| All participants | Poor      | 510  | 36.27 %                                            | 12.16 % | 42.94 % | 8.63 %    |
|                  | Rich      | 534  | 33.15 %                                            | 10.30 % | 49.44 % | 7.12 %    |
|                  | All       | 1044 | 34.67 %                                            | 11.21 % | 46.26 % | 7.85 %    |

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